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Herbivorous insects independently evolved salivary effectors to regulate plant immunity by destabilizing the malectin-LRR RLP NtRLP4

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eLife Assessment

This study provides an **important** contribution by showing that whiteflies and planthoppers use salivary effectors to suppress plant immunity through the receptor-like protein RLP4, suggesting convergent evolution in these insect lineages. The topic is of clear interest for understanding plant-insect interactions and offers ideas that could stimulate further research in the field. The authors provide mostly **solid** evidence for the functional roles of the salivary effectors; however, the interpretation of the physiological function of RLP4 in plant defense requires clarification.

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Abstract

Plants depend on receptor-like proteins and receptor-like kinases (RLPs/RLKs) to detect a variety of invading pathogens and insects. Although the mechanisms of microbial pathogens in suppressing plant immunity have been well investigated, it remains unclear how herbivorous insects overcome receptor-mediated defenses. Here, we show that salivary effectors evolve independently in whiteflies and planthoppers to target defensive RLP4, therefore minimizing the plant immunity. RLP4, as a leucine-rich repeat RLP (LRR-RLP), confers plant resistance against herbivorous insects by forming the RLP4/SOBIR1 complexes. In the whitefly *Bemisia tabaci*, BtRDP, the Aleyrodidae-specific salivary sheath protein, interacts with RLP4 from multiple plant species and promotes its ubiquitin-dependent degradation. Overexpression of NtRLP4 in transgenic plants exerts a detrimental impact on *B. tabaci* by exploiting the crosstalk between the salicylic acid (SA) and jasmonic acid (JA) pathways. Conversely, overexpression of BtRDP or silencing of NtRLP4 effectively alleviates such negative effects. In planthopper *Nilaparvata lugens*, the Delphacidae-restricted salivary protein NISP104 also targets and promotes the degradation of OsRLP4 from rice plants. Our results reveal the convergent evolution of salivary proteins in insects, and highlight the complex interactions between plants and herbivorous insects.

Introduction

Plants have evolved sophisticated defense systems to survive the constant attacks of pathogens and herbivorous insects. These defenses function at multiple levels, encompassing physical barriers such as the cuticle and cell wall, chemical defenses that include toxic secondary metabolites and anti-nutritive compounds, and indirect defenses that involve the emission of herbivore-induced volatiles to attract natural enemies of the herbivores [1, 2]. Beyond these

general strategies, plants also rely on the highly specialized molecular immune responses to detect and rapidly respond to invaders. The activation of innate immunity by membrane-localized receptors is a highly conserved mechanism among eukaryotes [3]. In sessile plants, the first line of defense against potential pathogenic microbes or insects relies on cell surface pattern-recognition receptors (PRRs) [4]. The plant genome encodes hundreds of PRRs, including receptor-like kinases (RLKs) and receptor-like proteins (RLPs) [4]. RLKs are consisted of different types of N-terminal ectodomains, a single transmembrane domain, and a cytoplasmic protein kinase domain, while RLPs have a similar structure but lack the kinase domain [5, 6]. As RLPs lack the intracellular signaling domains, they usually bind to SOBIR1 or SOBIR1-like adaptors to form the bimolecular receptor kinases [6]. Despite the large number of PRRs encoded by plant genomes, only a few of them have been well characterized regarding their functions in recognizing microbial-associated molecular patterns (MAMPs), damage-associated molecular patterns (DAMPs), and herbivore-associated molecular patterns (HAMPs) [5, 7]. For instance, FLS2 is known to recognize bacterial flagellin [8], EFR can recognize the bacterial elongation factor Tu [9], and EIX2 is responsible for the recognition of the fungal ethylene-inducing xylanase [10].

Piercing-sucking insects, such as whiteflies, aphids, and planthoppers, damage plants by feeding or transmitting viruses. They inject a mixture of saliva into plant tissues, which enhances their feeding efficiency by binding to leaked calcium, degrading extracellular DNA, and regulating plant hormonal signals [11–14]. In response, plants have evolved the ability to detect herbivores by sensing HAMPs present in insect saliva [7, 15]. This triggers pattern-triggered immunity (PTI), including mitogen-activated protein kinase (MAPK) cascades, reactive oxygen species (ROS), and hormone signaling [16–18]. A few salivary proteins have been reported to activate plant PTI, but the specific PRRs that recognize these proteins and trigger plant immunity are largely unclear [19–21]. One exception is the inceptin receptor (INR), which is a legume-specific LRR-RLP that can recognize the Vu-In in insect oral secretions [7]. Moreover, plant PRRs have been extensively reported to be targeted by pathogen effectors [22–24]. However, it remains unknown how insects cope with the immune recognition system.

The whitefly *Bemisia tabaci* (Hemiptera: Aleyrodidae) and brown planthopper *Nilaparvata lugens* (Hemiptera: Delphacidae) are notorious pests globally. During their feeding process, both insects secrete abundant species-specific salivary proteins into host plants [25, 26]. It is reported that plants initiate defense responses when they recognize insect feeding [19, 27]. However, the specific PRRs responsible for this process are still unknown. Both insects can successfully ingest phloem saps with limited plant defenses. At present, it remains unknown how insects attenuate plant defenses. Given that numerous defense mechanisms are conserved across plants [3, 28], it is interesting to investigate whether insect evolves a similar strategy to aid feeding during convergent evolution. In this study, the functions of RLPs in the plant immunity responsive to piercing-sucking insects were investigated, and *B. tabaci* and *N. lugens* were demonstrated to independently evolve salivary proteins to regulate the post-translational modification of RLPs.

Results

BtRDP is most abundantly expressed in salivary glands and secreted into host plants

Based on the transcriptomic results, BtRDP (*B. tabaci* RLP-degrading protein; GenBank accession: MN738093) was the most abundant gene in salivary glands [26]. It was a secretory protein that contained a signal peptide at the N-terminal, and totally 6 unique peptides from BtRDP were detected in the *B. tabaci* watery saliva (Fig. S1 [□](#)). As there was no gene homolog identified in the NCBI nr database, BtRDP was subjected to BLAST analysis against the transcriptomic or genomic data of 30 insect species. RDP was distributed in all the analyzed Aleyrodidae species, while no RDP homolog was detected in other species (Table S1). Phylogenetic analysis indicated that RDPs from five *B. tabaci* cryptic species were clustered in the same clade, exhibiting a close correlation with RDP from *Aleurocanthus spiniferus* (Fig. S2 [□](#)).

A group of *B. tabaci* were allowed to feed on *Nicotiana tabacum* plants for 24 h, and the presence of BtRDP in the infested and non-infested samples were assayed. One band between 26–33 kDa was detected in the *N. tabacum* plants infested by *B. tabaci*, but not in the non-infested plants (Fig. 1a). This band matched the molecular weight of BtRDP identified in the salivary glands. The detection of BtRDP in the infested plants (Fig. 1a) and in watery saliva (Fig. S1) collectively indicated that BtRDP was a salivary protein. According to Western-blotting, qPCR and transcriptomic analysis, BtRDP was nearly exclusively expressed in the salivary glands (Fig. 1b, c; Fig. S3a), which was significantly different from the actin or ribosomal 18S rRNA (Fig. S4a). In addition, BtRDP was highly expressed in the adult stages, but lowly expressed in the egg, nymph, and pseudopupa stages (Fig. 1d; Fig. S3b), similar to another salivary protein BtFTSP [29] but different from actin and ribosomal 18S rRNA (Fig. S4b). According to immunohistochemical (IHC) staining, BtRDP was specifically distributed in the principle salivary gland, but not in the accessory salivary gland or other tissues (Fig. 1e). For most piercing-sucking insects, two types of saliva (gel and watery saliva) are secreted into plant tissues. The salivary sheath, formed from gel saliva, serves as an insoluble lining along the stylet path, potentially providing a scaffold for effector delivery [30]. The salivary sheath secreted by *B. tabaci* was also stained with the anti-BtRDP serum. The BtRDP signal was mainly detected in the protuberant structure of sheath (Fig. 1f), suggesting that BtRDP attaches to salivary sheath after secretion.

BtRDP is important for *B. tabaci* feeding on tobacco plants

RNA interference was conducted to investigate the role of BtRDP in whitefly feeding. Treatment with dsBtRDP efficiently and specifically suppressed the target gene at both the transcript and protein levels (Fig. 2a; Fig. S5). There was no significant difference in the morphology of the salivary sheath between dsBtRDP-treated and dsGFP-treated *B. tabaci* (Fig. S6a). However, the salivary sheath length of dsBtRDP-treated *B. tabaci* was significantly shorter than that of the dsGFP-treated control (Fig. S6b). Furthermore, treatment with dsBtRDP did not affect the survivorship of whiteflies (Fig. S6c). Upon conducting a reproduction analysis, it was found that the dsBtRDP-treated whiteflies oviposited fewer eggs than the control (Fig. 2b). The feeding behavior of whiteflies was monitored using the electrical penetration graph (EPG) technique, which distinguishes nonpenetration (np), pathway (C), phloem salivation (E1), and phloem ingestion (E2) phases [31] (Fig. S7). Compared to the dsGFP control, dsBtRDP-treated *B. tabaci* exhibited a significant reduction in phloem ingestion and an extended pathway duration, indicating that BtRDP is necessary for efficient feeding (Fig. 2c).

Salivary proteins are reported to function in intercellular space and/or extracellular space after being secreted into host plants [30, 32]. To investigate the role of BtRDP in different subcellular locations of host plants, we constructed two transgenic *N. tabacum* lines overexpressing BtRDP: one carrying the full-length coding sequence that included the signal peptide (oeBtRDP), which is expected to be secreted into the apoplast (extracellular space), and the other lacking the signal peptide (oeBtRDP^{sp}), which is likely to be retained in the cytoplasm. The flag tag was fused to the C-terminal ends of the recombinant proteins, respectively. Transgenic plants containing the empty vector (EV), which exerted no significant influence on insect performance compared with the WT (Fig. S8), were used as negative controls. Fecundity and two-choice assays demonstrated that BtRDP, whether localized in the apoplast (oeBtRDP) or cytoplasm (oeBtRDP^{sp}), enhanced whitefly settling and oviposition relative to the EV controls (Fig. 2d–i; Fig. S9), indicating that BtRDP promotes whitefly feeding behavior irrespective of its subcellular location.

BtRDP interacts with the NtRPL4/NtSORBIR1 complex

To investigate the underlying mechanism of BtRDP in improving whitefly feeding, yeast two-hybrid (Y2H) screening against a *N. benthamiana* cDNA library was conducted using BtRDP^{sp} as a bait. Totally seven proteins were found to potentially interact with BtRDP^{sp} (Table S2), including an NbRPL4 (accession Niben261Chr07g1310001.1 in Niben261 genome). NbRPL4 possesses a

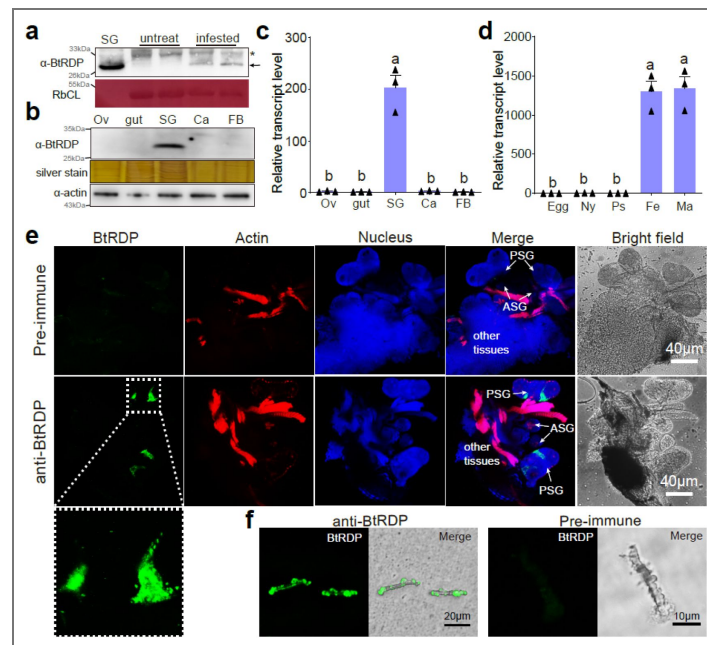


Fig. 1. BtRDP is a salivary protein and secreted into plants.

(a) Detection of BtRDP in *Nicotiana tabacum* plants. The untreated and *Bemisia tabaci*-infested tobacco plants, as well as the salivary gland samples, are collected for western-blotting assays. Asterisk indicates non-specific binding. Rubisco staining (RbCL) is conducted to visualize the amount of sample loading. (b, c) Detection of BtRDP in different tissues by western-blotting and qRT-PCR assays. Ov, ovary; SG, salivary gland; Ca, carcass; FB, fat body. Silver staining and anti-actin serum are used to visualize the sample loading. (d) Expression patterns of BtRDP in different development stages. Ny, nymph; Ps, pseudopupa; Fe, female; Ma, male. *B. tabaci tubulin* is used as an internal control. The relative quantitative method ($2^{-\Delta\Delta Ct}$) is used to evaluate the quantitative variation. Data are presented as mean values $\pm$ SEM ($n = 3$ independent biological replicates). Different lowercase letters indicate statistically significant differences at the $P < 0.05$ level according to one-way ANOVA test followed by Tukey's multiple comparisons test. (e, f) Immunohistochemical staining of BtRDP in salivary glands (e) and salivary sheath secreted from *B. tabaci* (f). The salivary gland and its nearby tissues are dissected and incubate with anti-BtRDP serum or pre-immune serum conjugated with Alexa Fluor™ 488 NHS Ester (green) and actin dye phalloidinrhodamine (red). The nucleus is stained with DAPI (blue). PSG, principal salivary gland; ASG, accessory salivary gland. The lower images represent the enlarged images of the boxed area in the upper image. Experiments are repeated thrice for (a and b), while twice for (e and f). Similar results are observed and representative images are displayed.

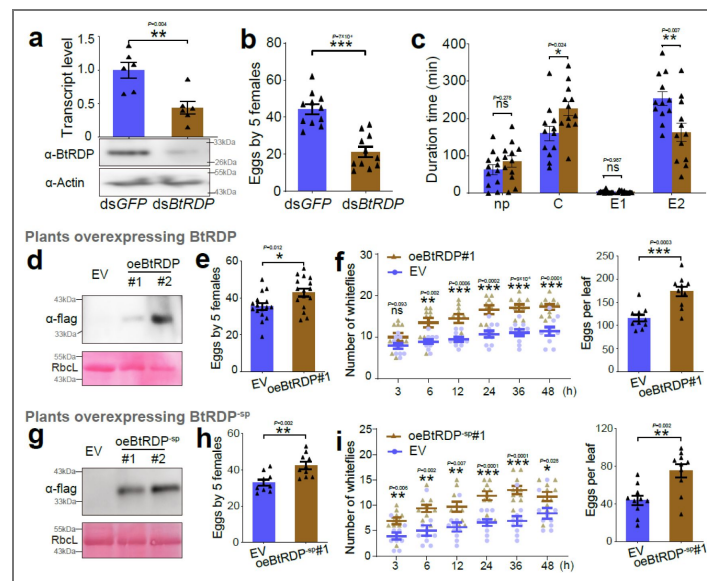


Fig. 2. Effects of BtRDP on *Bemisia tabaci*.

(a) Treating *B. tabaci* with dsBtRDP significantly reduces the transcript and protein level of target gene. The dsGFP-treated *B. tabaci* is used as a control. (b–c) Effects of *BtRDP* knockdown on insect reproduction (b) and feeding behavior (c). Electrical penetration graph (EPG) is used to monitor the insect feeding behavior, which can be classified into nonpenetration (np), pathway duration (C), phloem salivation (E1) and phloem ingestion (E2) phases. All EPG recordings are performed for 8 h. Typical EPG waveforms are displayed in Fig. S7. (d–f) Insect performance on transgenic plants overexpressing BtRDP (oeBtRDP). (d) Detection of BtRDP in transgenic *Nicotiana tabacum* overexpressing a complete coding region of *BtRDP*. The empty vector (EV) plants are used as the control. The flag tag is fused to the C-terminal ends of recombinant proteins. (e) Comparison of insect reproduction on oeBtRDP#1 and EV plants. Five *B. tabaci* individuals are confined to indicated plants for 3 days, and the oviposited eggs are counted. (f) Attraction of oeBtRDP#1 and EV leaves to *B. tabaci* in a two-choice equipment. A group of 40 female *B. tabaci* are released into a device containing oeBtRDP#1 and EV leaves. The number of insects settling on each leaf is counted. After 48 h, the number of eggs on each leaf is counted. (g–i) Insect performance on transgenic plants overexpressing BtRDP^{SP} (oeBtRDP^{SP}). (g) Detection of BtRDP^{SP} in transgenic *N. tabacum* overexpressing BtRDP without a signal peptide. The insect reproduction (h) and settlement (i) on oeBtRDP^{SP} transgenic plants are recorded. The EPG data are first checked for normality and homogeneity of variance, and data not fitting a normal distribution are subjected to log10 transformation. Data are presented as mean ± SEM. Statistical significance in panels (a), (b), (c), (e), (f), (h), and (i) was assessed using two-tailed unpaired Student's *t*-tests. ****P* < 0.001; ***P* < 0.01; **P* < 0.05; ns, not significant. Western-blotting assays are repeated thrice with the similar results.

predicted N-terminal signal peptide, a leactin-like domain, a LRR domain, and a transmembrane domain. NbRLP4, as well as its closest homology in *N. tabacum* (NtRLP4; GenBank accession: XM_016601109), did not contain the kinase domain (Fig. 3a).

The relative expression of *NtRLP4* in response to *B. tabaci* infestation was investigated. Based on qRT-PCR results, the transcript level of *NtRLP4* was not significantly influenced by *B. tabaci* (Fig. S10). However, a decrease in the protein level of NtRLP4 was detected by Western-blotting assay (Fig. 3b). Point-to-point Y2H assays revealed that NtRLP4₍₂₃₋₅₄₁₎ (a truncated version lacking the signal peptide and transmembrane domains) interacted with BtRDP^{SP} (Fig. 3c). To investigate the specificity of the NtRLP4-BtRDP interaction, another RLP protein NtCf9 with the highest sequence similarity to tomato resistance protein Cf9 (Fig. S11) [33], as well as three randomly selected salivary proteins from *B. tabaci* (BtFTSP, QHB15613; BtSP16.3, MN738005; BtSP37.4, MN738158), were used as controls. As a result, BtRDP^{SP} did not interact with the truncated NtCf9 construct, in which the signal peptide and transmembrane domains were removed (Fig. 3c). Also, NtRLP4₍₂₃₋₅₄₁₎ could not interact with the other three salivary proteins without signal peptides (Fig. S12a, b). These findings raised the question of which domain of NtRLP4 is responsible for binding to BtRDP, as identifying the interacting domain can help to infer where the salivary protein contacts the receptor in plants. We therefore dissected the NtRLP4 domains accordingly. The results showed that it was the LRR domain, rather than the leactin-like domain or the short intercellular region, that was required for the NtRLP4-BtRDP interaction (Fig. 3c). To investigate the shortest LRR sequence responsible for BtRDP interaction, the LRR domain was divided into three segments. However, none of these segments interacted with BtRDP^{SP} (Fig. S12a). Co-immunoprecipitation (Co-IP) assays further confirmed the specific interaction between BtRDP and NtRLP4 (Fig. 3d). It was the C-terminal myc tag-fused NtRLP4 (NtRLP4-myc), but not the NtCf9 (NtCf9-myc), that could be immunoprecipitated by the C-terminal flag tag fused BtRDP (BtRDP-flag). In the bimolecular fluorescence complementation (BiFC) assay, YFP fluorescence was observed upon co-expression of the N-terminal nYFP tag fused NtRLP4 and N-terminal cYFP tag fused BtRDP^{SP} (Fig. S12c). Furthermore, in oeBtRDP transgenic plants, endogenous NtRLP4 was specifically immunoprecipitated with BtRDP-flag, which was significantly different from the control (Fig. S12d). These results suggest that BtRDP interacts with NtRLP4. The absence of kinase motifs in the C-terminal region of NtRLP4 suggests that it may function in cooperation with other proteins to initiate signaling. The receptor-like kinase SOBIR1, which contained a kinase domain (Fig. S13a), has been widely reported to be required for the function of RLPs in the innate immunity [6]. Through Co-IP assays, our study demonstrated that NtSOBIR1 was capable of interacting with NtRLP4 but not BtRDP (Fig. S13b, c).

NtRLP4 and NtSOBIR1 are resistant to *B. tabaci* infestation

The transgenic *N. tabacum* overexpressing NtRLP4 (oeRLP#1 and oeRLP#2, without any tags) was constructed (Fig. 3e). Compared with the EV control, both oeRLP lines were less attractive to *B. tabaci*, alongside less settling and ovipositing insects (Fig. 3f; Fig. S14a). Also, *B. tabaci* had a significantly decreased fecundity in oeRLP plants (Fig. 3g; Fig. S14b). Similar results were observed after transient overexpression of NtRLP4-GFP in *N. tabacum* (Fig. S14c, d), indicating that NtRLP4 is resistant to *B. tabaci* infestation.

Thereafter, the EV and oeRLP#1 plants underwent transcriptomic sequencing (Fig. S15a). A total of 729 differentially expressed genes (DEGs) were identified, comprising 423 up-regulated and 306 down-regulated genes (Table S3). Enrichment analysis demonstrated that the majority of up-regulated DEGs were associated with plant-pathogen interaction, environmental adaptation, MAPK signaling pathway, and signal transduction (Fig. S15b). In contrast, the glutathione metabolism, carbohydrate metabolism, and amino acid metabolism pathways were significantly enriched in down-regulated DEGs (Fig. S15c). Noteworthy, numerous DEGs were annotated as RLK/RLP or WRKY transcription factor, with the majority of them being significantly upregulated (Fig. S15d, e). These results suggested an enhanced defense response in oeRLP plants. The crosstalk between salicylic acid (SA) and jasmonic acid (JA) pathways was critical for plant defenses [14]. The increased defense of oeRLP transgenic plants was partially caused by the

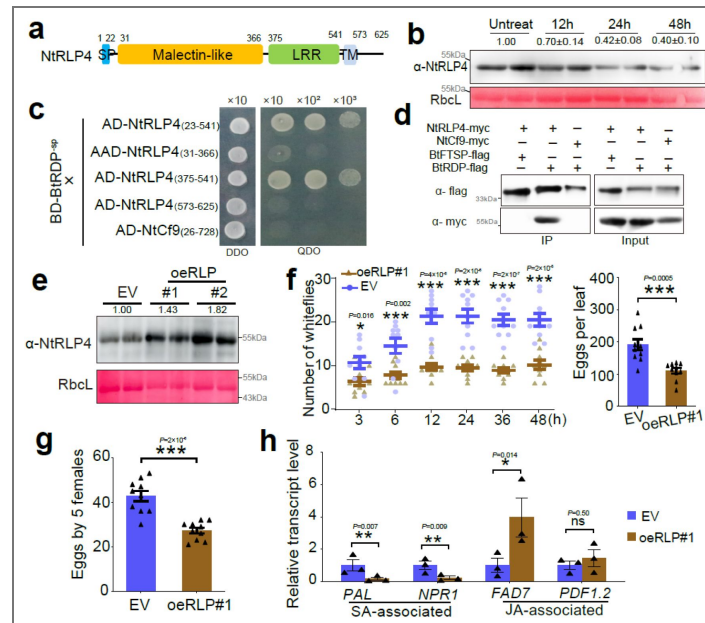


Fig. 3. NtRPL4 interacts with BtRDP and confers plant resistance to *Bemisia tabaci*.

(a) Domain organization of NtRPL4. NtRPL4 contains a predicted N-terminal signal peptide (SP), a malectin-like domain, a LRR domain, and a transmembrane domain (TM). (b) The protein level of NtRPL4 in response to *B. tabaci* infestation. The untreated *Nicotiana tabacum* is used as a negative control. Four independent biological replicates are performed, and the representative images are displayed. The band density is measured using ImageJ. The density values from 4 biological replicates are calculated and the mean value in the controls is set at 1.0. (c) Yeast two hybrid assays showing the interaction between BtRDP and NtRPL4. The different combinations of constructs are transformed into yeast cells, and are grown on the selective medium SD/-Trp/-Leu (DDO), and the interactions are tested with SD/-Trp/-Leu/-His/-Ade (QDO). (d) Co-immunoprecipitation assay showing the interaction between BtRDP and NtRPL4. Total proteins are extracted from *N. benthamiana* leaves transiently co-expressing NtRPL4-myc/NtCf9-myc with BtFTSP-flag/BtRDP-flag. All genes are expressed with a complete coding region, and the myc/flag tags are fused at the C-terminus. Precipitation is performed using flag beads. The samples are probed with anti-flag and anti-myc antibodies for immunoblotting analysis. (e-h) Analysis of transgenic *N. tabacum* overexpressing NtRPL4 (oeRPL). (e) Detection of NtRPL4 level in oeRPL plants. The empty vector (EV) plant is used as a control. Two independent oeRPL lines are selected. The samples are probed with an anti-NtRPL4 antibody. Rubisco staining (RbcL) is used to visualize the amount of sample loading. (f) Attraction of oeRPL#1 and EV leaves to *B. tabaci* in a two-choice equipment. A group of 40 female *B. tabaci* are released into a device containing oeRPL#1 and EV leaves. The number of insects settling on each leaf is counted at each time point. After 48 h, the number of eggs on each leaf is counted. (g) Comparison of insect reproduction on oeRPL#1 and EV plants. Five *B. tabaci* individuals are confined to indicated plants for 3 days, and the oviposited eggs are counted. (h) Relative transcript levels of salicylic acid (SA)- and jasmonic acid (JA)-associated genes in oeRPL#1 and EV plants. PAL, phenylalanine ammonia lyase; NPR1, nonexpressor of pathogenesis-related protein 1; FAD7, fatty acid desaturase 7; PDF1.2, plant defensin 1.2. Two independent biological replicates are performed in (d) and (e). Data in (f), (g), and (h) are presented as mean values ± SEM. For insect bioassays in (f) and (g), n = 10 independent biological replicates. For qRT-PCR in (h), n = 3 independent biological replicates. P-values are determined by two-tailed unpaired Student's *t* test. ****P* < 0.001; ***P* < 0.01; **P* < 0.05; ns, not significant.

altered hormonal signaling. The JA-associated gene *FAD7* was significantly induced, while the SA-associated genes *PAL* and *NPR1* were repressed in oeRLP plants (Fig. 3h). Transient overexpression of NtRLP4-GFP in *N. tabacum* plants via *Agrobacterium* infiltration exerted similar effects (Fig. S14e). Furthermore, NtRLP4-GFP significantly enhanced H₂O₂ accumulation (Fig. S14f).

The function of NtSOBIR1 was also investigated by transient overexpression and hairpin RNAi (Fig. S16). Overexpression of NtSOBIR1-GFP induced the cell death phenotype five days post agro-injection (Fig. S16). Insects preferred to settle and oviposit on GFP-expressed *N. tabacum* than on NtSOBIR1-GFP-expressed control, even though no obvious death phenotype was observed two days post agro-injection (Fig. S16c, d). For NtSOBIR1 silencing, no cell death phenotype was observed, while the NtSOBIR1-silenced leaves exhibited reduced growth (Fig. S16f). Compared with the EV-silenced control, NtSOBIR1-silenced *N. tabacum* was slightly attractive to *B. tabaci* (Fig. S16g, h). Collectively, these results demonstrate that NtSOBIR1 is resistant to *B. tabaci* infestation.

BtRDP attenuates plant defenses by promoting the degradation of host RLP4

To investigate the interplay between NtRLP4 and BtRDP, transgenic *N. tabacum* silencing NtRLP4 (*RNAi-RLP#1* and *RNAi-RLP#2*) plants were constructed. Both *RNAi-RLP* lines showed reduced NtRLP4 levels compared with EV plants, with *RNAi-RLP#2* exhibiting a stronger silencing effect (Fig. S17a). Compared with EV plants, *B. tabaci* showed a tendency to settle on NtRLP4-silenced plants and lay more eggs (Fig. S17b, c). Thereafter, dsGFP- and dsBtRDP-treated *B. tabaci* were reared on EV, *RNAi-RLP#1*, and *RNAi-RLP#2*, respectively. The dsBtRDP-treated *B. tabaci* produced less offspring than the control (decreased by 28.3%; $P=0.0007$) on EV plants (Fig. 4a), consistent with the results obtained from WT plants (Fig. 2b). On the *RNAi-RLP#1* plant, the impaired insect reproduction was still observed after dsBtRDP treatment, but the extent was slightly reduced (decreased by 23.7%; $P=0.002$) (Fig. 4a). On the *RNAi-RLP#2* plant, there was no significant difference between the dsGFP and dsBtRDP treatments (Fig. 4a). The differential rescue effect between the two RNAi lines likely resulted from their different NtRLP4 silencing efficiencies, with the lower NtRLP4 level in *RNAi-RLP#2* leading to a more complete rescue phenotype (Fig. S17a; Fig. 4a). Collectively, these results suggest that *RNAi-RLP* plants are beneficial for insects and can partially, but not completely, rescue the impaired feeding performance of dsBtRDP-treated insects.

To investigate the influence of BtRDP on NtRLP4, GFP-tagged NtRLP4 (NtRLP4-GFP) was transiently co-expressed with mCherry-tagged BtRDP (BtRDP-mCherry), red fluorescent protein (RFP)-mCherry, and BtFTSP1-mCherry through agroinfiltration, respectively. All tags were fused with the C-terminal region. As discovered, BtRDP-mCherry did not influence the location patterns of NtRLP4-GFP (Fig. S18). However, a significant reduction in the fluorescent signal of NtRLP4-GFP was detected after co-expression with BtRDP-mCherry, but not with the controls (Fig. S18). Besides, co-expression with BtRDP-mCherry/NtRLP4-GFP significantly reduced H₂O₂ accumulation (Fig. 4b), suppressed the expression of JA-associated genes *FAD7* and *PDF.12*, and induced that of the SA-associated genes *PAL* and *NPR1* (Fig. 4c). It was discovered by insect bioassays that, transient overexpression of BtRDP-mCherry rescued the negative effects on whitefly performance caused by NtRLP4-GFP overexpression (Fig. 4d, e). In the host choice experiments, *B. tabaci* displayed a preference for settling on *N. tabacum* leaves co-expressing BtRDP-mCherry/NtRLP4-GFP over those co-expressing RFP-mCherry/NtRLP4-GFP (Fig. 4d). However, almost no preference was observed in *B. tabaci* feeding on *N. tabacum* plants co-expressing BtRDP-mCherry/NtRLP4-GFP and plants expressing GFP alone (Fig. S19).

The attenuated NtRLP4-GFP fluorescent signal (Fig. S18), along with the observed decrease in the protein level of NtRLP4 during *B. tabaci* infestation (Fig. 3b), led us to speculate on whether BtRDP influenced the accumulation of NtRLP4. To this end, the BtRDP^{SP}-his and GFP-his proteins were prokaryotically expressed in *Escherichia coli*. Different concentrations of purified BtRDP^{SP}-his and GFP-his proteins were infiltrated into *N. tabacum* plants overexpressing NtRLP4-myc.

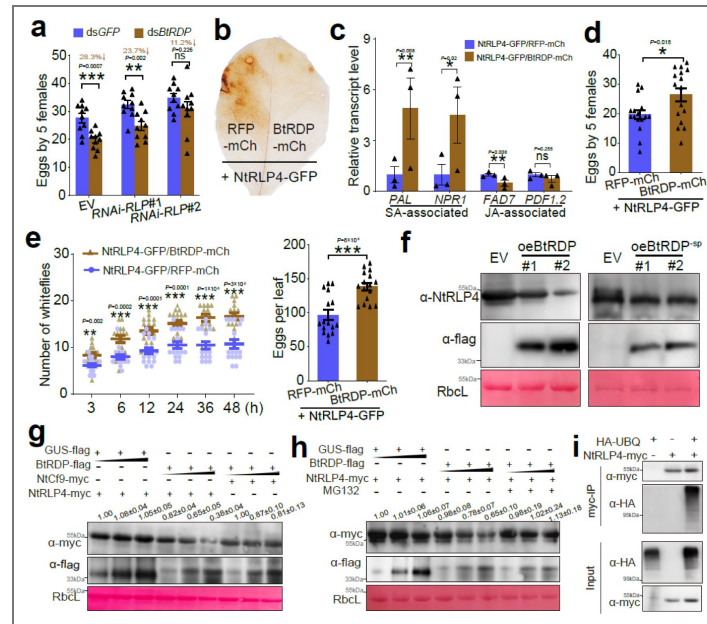


Fig. 4. BtRDP suppress plant defenses by promoting NtRPL4 degradation.

(a) Effects of dsBtRDP suppression on insect reproduction ($n = 10$ independent biological replicates) when feeding on empty vector (EV) and NtRPL4-silenced (RNAi-RPL) transgenic *Nicotiana tabacum*. (b) Transient overexpressing BtRDP-mCherry attenuates H_2O_2 accumulation caused by NtRPL4-GFP overexpression. *N. tabacum* co-expressing RFP-mCherry and NtRPL4-GFP is used as a control. All genes are expressed with a complete coding region. The experiment is repeated 5 times with the similar results. (c) Relative transcript level of salicylic acid (SA)- and jasmonic acid (JA)-associated genes in BtRDP-mCherry/NtRPL4-GFP and RFP-mCherry/NtRPL4-GFP plants ($n = 3$ independent biological replicates). PAL, phenylalanine ammonia lyase; NPR1, nonexpressor of pathogenesis-related protein 1; FAD7, fatty acid desaturase 7; PDF1.2, plant defensin 1.2. (d) Comparison of insect reproduction ($n = 16$ independent biological replicates) on transiently-expressed BtRDP-mCherry/NtRPL4-GFP and RFP-mCherry/NtRPL4-GFP plants. (e) Attraction of BtRDP-mCherry/NtRPL4-GFP and RFP-mCherry/NtRPL4-GFP leaves to *B. tabaci* in a two-choice equipment ($n = 16$ independent biological replicates). The number of eggs on each leaf is counted at 48 h post insect release. (f) Effects of BtRDP and BtRDP^{SP} overexpression on NtRPL4 accumulation. Transgenic oeBtRDP (left) and oeBtRDP^{SP} (right) plants are probed with anti-NtRPL4 and anti-flag antibodies for immunoblotting analysis. The experiments are repeated twice with the similar results. (g) Degradation of NtRPL4 by BtRDP in *N. benthamiana* leaves. NtRPL4-myc and BtRDP-flag are transiently co-expressed in *N. benthamiana* plants through *Agrobacterium* infiltration. *Agrobacterium* carrying NtCf9-myc and GUS-flag are used as negative controls. (h) Effects of 26S proteasome inhibitor (MG132) on NtRPL4 accumulation. Co-infiltrated leaves are treated with MG132 at 24 h post injection. The samples are probed with anti-flag and anti-myc antibodies for immunoblot analysis. Rubisco staining (Rbcl) is conducted to visualize the amount of sample loading. The small triangle indicates the different concentrations of *Agrobacterium* ($OD_{600} = 0.05, 0.3, \text{ and } 1.0$). (i) NtRPL4 is ubiquitinated *in planta*. NtRPL4-myc is co-expressed transiently with HA-UBQ in *N. benthamiana* leaves. Extracted total proteins are immunoprecipitated by anti-myc beads and immunoblotted with anti-myc or anti-HA antibody. Experiments are repeated thrice for (g) and (h), while twice for (i). Band density is measured using ImageJ. The density values from three biological replicates are calculated and the mean value in the first lane is set at 1.0. Data are presented as mean values $\pm$ SEM. *P*-values are determined by two-tailed unpaired Student's *t* test. ****P* < 0.001; ***P* < 0.01; **P* < 0.05; ns, not significant.

Interestingly, the protein level of NtRPL4-myc slightly decreased with the increasing amount of BtRDP^{SP}-his, while no such effect was observed with GFP-his (Fig. S20 [Fig. S20](#)). Moreover, the BtRDP^{SP}-his protein did not have any significant effect on the accumulation of another RLP NtCf9-like (Fig. S20 [Fig. S20](#)). We further assessed the impact of BtRDP on NtSOBIR1 following NtRPL4 destabilization. As the amount of BtRDP-flag increased, NtRPL4-myc accumulation was markedly reduced, while NtSOBIR1-flag levels remained unchanged (Fig. S21 [Fig. S21](#)).

Then, the NtRPL4 levels in oeBtRDP and oeBtRDP^{SP} plants were detected. Compared with the EV control, a reduced NtRPL4 abundance was observed in both oeBtRDP#1 and oeBtRDP#2 transgenic lines, but not the oeBtRDP^{SP} plants (Fig. 4f [Fig. 4f](#)), suggesting the importance of BtRDP secretion in extracellular space. BtRDP attached to salivary sheath that was localized in plant apoplast (Fig. 1f [Fig. 1f](#)). The presence of signal peptide promoted the secretion of BtRDP into the extracellular space. Therefore, the complete coding region of *BtRDP* was used in subsequent experiments. The NtRPL4-myc and BtRDP-flag were transiently co-expressed in *N. benthamiana* plants through *Agrobacterium* infiltration. As a result, BtRDP-flag significantly decreased the protein level of NtRPL4-myc, but not NtCf9-myc (Fig. 4g [Fig. 4g](#)). qRT-PCR analyses demonstrated that the transcript level of *NtRPL4* remained unchanged (Fig. S22 [Fig. S22](#)). These findings indicate that the reduction in the protein level of NtRPL4 is not caused by transcriptional changes but rather by degradation.

The ubiquitin system and autophagy are two major pathways regulating protein degradation [34]. At first, MG132, a 26S proteasome inhibitor known to inhibit the degradation of ubiquitin-conjugated proteins, was employed [35]. The results showed that MG132 efficiently prevented the degradation of NtRPL4-myc by BtRDP-flag (Fig. 4h [Fig. 4h](#)). Then, the autophagy inhibitor 3-Methyladenine (3-MA) and bafilomycin A1 (BAF) that can block autophagy in plants were used [36]. However, the decreased protein level of NtRPL4-myc was still observed in the presence of 3-MA or BAF (Fig. S23 [Fig. S23](#)). Additionally, NtRPL4-myc and HA-tagged ubiquitin (HA-UBQ) were co-expressed in *N. benthamiana*. Total proteins were extracted from *N. benthamiana* leaves and then subjected to immunoprecipitation with anti-myc beads. The ubiquitin attached to NtRPL4 was determined by immunoblot using the anti-HA antibody. Leaves expressing NtRPL4-myc or UBQ-HA alone were used as negative controls. The results showed that a strong UBQ signal was detected in the immunoprecipitated product only when NtRPL4-myc and UBQ-HA were co-expressed (Fig. 4i [Fig. 4i](#)). The molecular weight of UBQ-attached NtRPL4-myc was remarkably larger than NtRPL4-myc itself (Fig. 4i [Fig. 4i](#)). These results suggest that BtRDP promotes NtRPL4 degradation through the ubiquitin system, but not the autophagy pathway.

Degrading plant RLP4s by salivary effectors is a common strategy for insects

To investigate the ability of BtRDP to degrade RLPs from other host plants of *B. tabaci*, *Solanum lycopersicum* RLP4 (SlRPL4; GenBank accession: XP_004232910) was selected for further analysis. Sequence analysis revealed a similar gene structure between SlRPL4 and NtRPL4 (Fig. S24 [Fig. S24](#); Fig. S25a [Fig. S25a](#)). Moreover, the interaction between BtRDP and SlRPL4 was confirmed through Y2H and Co-IP assays (Fig. S25b [Fig. S25b](#), c). Additionally, SlRPL4-myc and SlCf9-like-myc were transiently co-expressed with varying concentrations of BtRDP-flag or GUS-flag, with the myc and flag tags fused with the C-terminus. The results showed that BtRDP-flag exerted no influence on the accumulation of SlCf9-like-myc protein. In contrast, the protein level of SlRPL4-myc was significantly reduced in the presence of BtRDP-flag (Fig. S25d [Fig. S25d](#)), indicating the conserved function of BtRDP in *B. tabaci* adaptation to different host plants.

BtRDP was a gene specific to the Aleyrodidae family, lacking homologues in other species. To explore whether the degradation of RLPs by salivary effectors was a conserved mechanism among different herbivorous insects, a RLP4 homology from *Oryza sativa* (OsRPL4; GenBank accession: XP_015645303; Fig. 5a [Fig. 5a](#)) and twenty salivary proteins from planthopper species lacking signal peptides were paired to identify the potential interactions using Y2H assays. As a result, only the salivary protein NISP104^{SP} (*N. lugens* salivary protein consisting of 104 amino acids, GenBank accession: MF278694) interacted with OsRPL4₍₂₉₋₅₅₁₎ (OsRPL4 lacking signal peptides and transmembrane domains), while the other salivary proteins tested did not show any interaction

(Fig. 5b, c [Fig. S26](#)). NISP104 did not exhibit any sequence or structural similarity to BtRDP (Fig. S27 [Fig. S12a](#)). Truncation analysis revealed that NISP104^{SP} interacted with both the LRR domain and the malectin-like domain of OsRLP4 (Fig. 5c [Fig. S12a](#)), which was slightly different from the BtRDP-NtRLP4 interaction (Fig. 3c [Fig. S12a](#)). Given that both BtRDP and NISP104 interacted with the LRR domain of RLP4, further investigation focused on the specific region of the LRR domain responsible for NISP104 binding. The LRR domain of OsRLP4 was divided into three segments, mirroring the division of the LRR domain in NtRLP4 (Fig. S12a [Fig. S12a](#)). NISP104^{SP} interacted with the truncated LRR domain (OsRLP4₍₄₈₀₋₅₂₆₎) (Fig. 5c [Fig. S12a](#)). In contrast, none of the truncated LRR domain of NtRLP4 interacted with BtRDP^{SP} (Fig. S12a [Fig. S12a](#)). These findings suggest that the binding sites for salivary proteins to RLP4 may differ between *N. lugens* and *B. tabaci*.

NISP104 was a planthopper-specific salivary protein nearly exclusively expressed in salivary glands (Table S1; Fig. 5d [Fig. S28](#)). The function of NISP104 or its gene homologs in other planthopper species has not been well investigated previously. Silencing NISP104 did not affect insect survivorship (Fig. 5e [Fig. S28](#)). However, significant reductions in honeydew excretion and impaired fecundity were observed in dsNISP104-treated *N. lugens* (Fig. 5f, g [Fig. S28](#)), indicating the importance of NISP104 in insect feeding.

Subsequently, OsRLP4-myc was transiently co-expressed with a complete coding region of NISP104 (C-terminal flag tagged, NISP104-flag). Another salivary protein NISP7-flag and NtCf9-myc were used as negative controls. It was found that NISP104-flag specifically and significantly decreased the protein level of OsRLP4-myc, which differed significantly from that of the NISP7-flag (Fig. 5h [Fig. S28](#)). Additionally, NISP104-flag failed to decrease the protein level of NtCf9-myc (Fig. 5h [Fig. S28](#)). These findings suggest that *N. lugens* potentially evolves salivary effectors to target rice RLPs.

Discussion

Plants have evolved a complex innate immune system to protect themselves against pathogen/insect invasion, and the cell surface-localized PRRs function on the frontline [4]. To counteract this response, the pathogens have developed various strategies to avoid plant PRR-mediated PTI for their own advantage [37]. However, it remains unknown whether insects take the similar strategy to sabotage plant immunity. In this study, we demonstrate that the RLP4/SOBIR1 complexes confer plant resistance against herbivorous insects. In response, both planthoppers and whiteflies secrete salivary effectors that facilitate the degradation of defensive RLP4, leading to the attenuated plant immunity. Our results suggest that manipulation of PRRs may be a conserved strategy for herbivorous insects.

PRRs detect modified-self or non-self patterns and initiate a cascade of immune signaling events, which subsequently regulate downstream defense responses, conferring plant resistance against invaders [4, 5]. Although various insect salivary components have been identified as triggers for plant immune responses, the specific PRRs involved in this process remain unclear. Our study indicates that RLP4 significantly influences insect performance by modulating plant hormones and ROS (Fig. 3f-h [Fig. S14](#)), highlighting the critical role of RLP4 in plant immunity. However, the precise ligand for RLP4 has yet to be identified. The malectin-LRR receptor RLP4 may respond to the DAMPs or HAMPs released during insect feeding [38], a hypothesis that warrants further investigation. RLP4 lacks an intracellular kinase domain, and instead forms bimolecular receptor kinases by interacting with SOBIR1 (Fig. S13 [Fig. S13](#)). Overexpression of NtRLP4 or NtSOBIR1 in *N. tabacum* attracts *B. tabaci* and promotes insect reproduction, whereas silencing of either gene exerts the opposite effect (Fig. S14 [Fig. S17](#)). The association between RLPs and SOBIR1 protein kinases is structurally and functionally analogous to that of genuine receptor kinases [6]. Therefore, destabilizing the RLP4/SOBIR1 complex can potentially disrupt the immune recognition processes. It is noteworthy that although NtRLP4 interacts with SOBIR1, this alone does not confirm that it operates strictly through this canonical module. Evidence from other RLPs shows that co-receptor usage can be flexible, and some RLPs function partly or conditionally independent of SOBIR1 [39]. Therefore, a more definitive assessment of NtRLP4 signaling will therefore require genetic dissection of its co-receptor dependencies, including but not limited to SOBIR1.

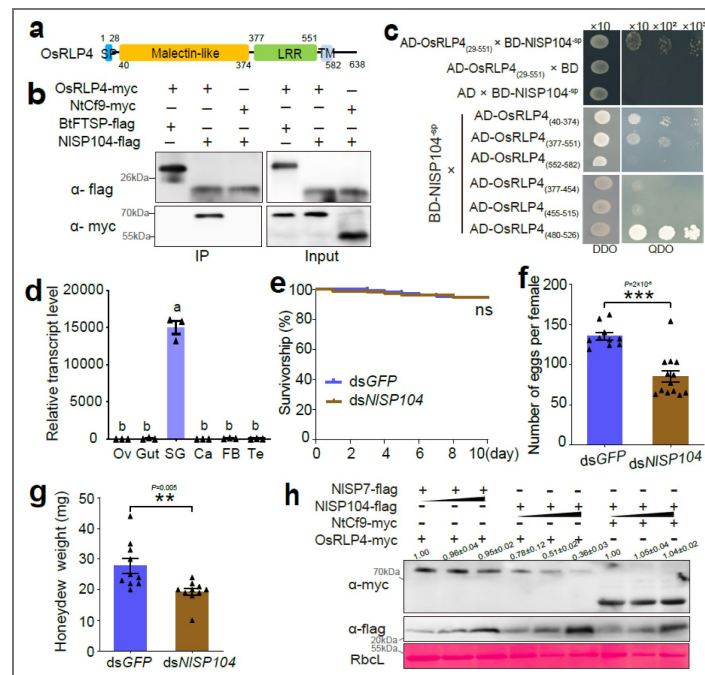


Fig. 5. Rice RLP4 are targeted by salivary protein NISP104 from *Nilaparvata lugens*.

(a) Domain organization of *Oryza sativa* RLP4 (OsRPL4). (b, c) Yeast two-hybrid and Co-IP assays showing the interaction between OsRPL4 and NISP104. All genes are expressed with a complete coding region in Co-IP assays, while NISP104 without a signal peptide is used in yeast two-hybrid assays. Experiments in (b) are repeated twice with the similar result. (d) The expression patterns of *NISP104* in different *N. lugens* tissues. Te, testis; Ov, ovary; SG, salivary gland; Ca, carcass; FB, fat body. Data are presented as mean values $\pm$ SEM ($n = 3$ independent biological replicates). Different lowercase letters indicate statistically significant differences at $P < 0.05$ level according to one-way ANOVA test followed by Tukey's multiple comparisons test. (e-g) Effects of dsRNA treatment on insect survivorship (e), reproduction (f), and honeydew excretion (g). For survivorship analysis, a group of 30 *N. lugens* are reared in a cage. Three independent biological replications are performed. Differences in survivorship between the two treatments are tested by log-rank test. ns, not significant. For reproduction analysis, $n = 10$ and 13 individuals are tested in dsGFP- and dsNISP104-treatment, respectively. The sterile females are excluded from data analysis. For honeydew analysis, $n = 10$ independent biological replicates. P -values in (f) and (g) are determined by two-tailed unpaired Student's t test. *** $P < 0.001$; ** $P < 0.01$. (h) Effect of NISP104 on the accumulation of transiently expressed OsRPL4. OsRPL4-myc is agro-injected into *Nicotiana benthamiana* together with different concentrations of NISP104-flag and NISP7-flag. NtCf9-myc is used as a negative control. The small triangle indicates the different concentrations ($OD_{600} = 0.05, 0.3$, and 1.0) of *Agrobacterium*. Rubisco staining (RbCL) is conducted to visualize the amount of sample loading. Experiments are repeated three times with the similar results. Band density is measured using ImageJ. The density values from three biological replicates are calculated with the mean value in the first lane being set at 1.0.

As pivotal regulators in immune signaling, PRRs are frequently targeted by multiple pathogens through diverse mechanisms. For example, *Pseudomonas syringae* hampers the functionality of receptor complexes by disrupting the ligand-dependent association between receptor-like kinase BAK1 and the flagellin receptor FLS2 [24], while *Phytophthora infestans* promotes alternative splicing of the host RLPs to enhance infection [23]. PRRs are usually subjected to degradation via post-translational modifications [40]. In plants, the activity and stability of PRRs are under stringent regulation since excessive activation of PRRs could be detrimental [22, 41]. One prevalent regulatory mechanism involves ubiquitination followed by subsequent degradation mediated by the 26S proteasome [40]. Mounting evidence has demonstrated that bacteria, fungi, viruses, and oomycetes exploit such mechanisms to facilitate PRR degradation [42–45]. Nevertheless, the occurrence of effector-mediated PRR degradation in insects is unreported. Our study reveals that the salivary effectors BtRDP from *B. tabaci* and NISP104 from *N. lugens* interact with plant RLP4s, resulting in their degradation in a ubiquitin-dependent manner (Fig. 4f–i).

Notably, BtRDP and NISP104 exhibit neither sequence nor structural similarity and do not resemble the known eukaryotic ubiquitin-ligase domains. Their interaction with RLP4s occurs in the extracellular space (Fig. 3d; Fig. 5c), whereas the ubiquitin-proteasome system primarily functions within the cytosol and nucleus [46]. Furthermore, NtRLP4 reduction is observed solely in oeBtRDP transgenic plants, not in oeBtRDP^{SP} plants (Fig. 4f), indicating that BtRDP exerts its influence on NtRLP4 in the extracellular space. These observations collectively refute the possibility that BtRDP or NISP104 possesses the intrinsic E3 ligase activity capable of directly ubiquitinating RLP4s within plant cells. Importantly, the reduced NtRLP4 levels may not stem from a direct physical interaction between BtRDP and NtRLP4. Instead, BtRDP may indirectly affect RLP4 post-translational modification, thereby accelerating its degradation, which warrants further investigation. Additionally, the independent evolution of RLP4-targeting effectors in various insect lineages may expedite plant-insect co-evolution. Plants with robust RLP4-mediated defenses can select for herbivores carrying effective effectors, driving an evolutionary “arms race”. Conversely, insects targeting similar host proteins may exert parallel selective pressures on unrelated plant species, potentially shaping defense diversification across plant communities. Understanding these dynamics provides insight into how molecular interactions scale up to influence herbivore adaptation, host range, and the ecological outcomes of plant-insect interactions.

Herbivorous insects employ a mix of salivary components to ensure successful feeding [30]. Previous research has indicated that *B. tabaci* suppresses plant defenses by utilizing SA-JA crosstalk, with the salivary protein Bt56 playing a crucial role in this process [14]. Bt56 activates the SA-signaling pathway by directly interacting with a KNOTTED 1-like homeobox transcription factor [14]. This study investigates the effects of another salivary effector, BtRDP, on plant hormones, and finds that it elicits similar effects to Bt56, albeit primarily impacting the signal recognition process (Fig. 3h; Fig. S14e). This leads us to speculate that different salivary effectors from the same insect species may simultaneously target the same signaling pathway to ensure effective immune suppression. Our analysis reveals that BtRDP interacts with other host genes apart from NtRLP4 (Table S2), and silencing NtRLP4 cannot completely rescue the impaired feeding performance of dsBtRDP-treated insects (Fig. 4a). Salivary proteins exert a crucial effect on mediating the interactions between herbivores and plants. The dynamic interplay between plants and insects may be highly complex, one salivary protein can target multiple plant pathways, and one plant pathway is influenced by multiple salivary proteins.

In conclusion, this study shows that suppressing PRR-mediated plant immunity may be the common strategy for the successful feeding of herbivorous insects. Whiteflies and planthoppers independently evolve salivary effectors that facilitate the degradation of defensive RLP4 in host plants in an ubiquitin-dependent manner, therefore disrupting the RLP4/SOBIR1 complexes for signal recognition (Fig. 6).

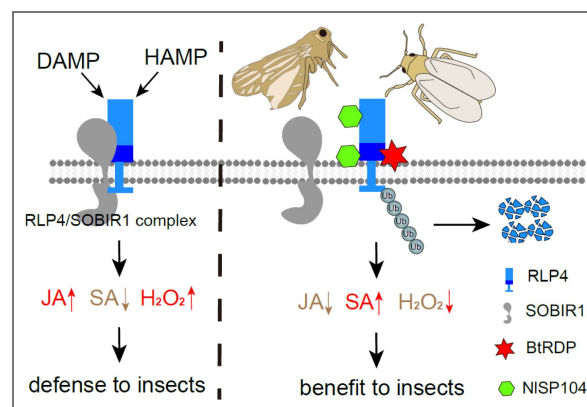


Fig. 6. The proposed model for the suppression of receptor-like proteins (RLPs)-mediated plant defenses by salivary effectors.

Host plants employ pattern-recognition receptors (PRRs) to detect various damage-associated molecular patterns (DAMPs) and herbivore-associated molecular patterns (HAMPs) triggered by insect feeding. The RLP4/SOBIR1 complex plays a vital role in initiating pattern-triggered immunity (PTI), including H_2O_2 burst, upregulation of jasmonic acid (JA), and downregulation of salicylic acid (SA), which hinders insect feeding. The whitefly *Bemisia tabaci* and planthopper *Nilaparvata lugens* independently evolved salivary proteins that targeted plant RLP4. *B. tabaci* salivary sheath protein BtRDP interacts with the leucine-rich repeat (LRR) domain of RLP4 from *Nicotiana tabaci* and *Solanum lycopersicum*, while *N. lugens* NISP104 targets both the LRR domain and the malectin-like domain of *Oryza sativa* RLP4. These interactions promote the ubiquitin-dependent degradation of RLP4, thereby disrupting the stability of the RLP4/SOBIR1 complex. The presence of salivary effectors causes a hormonal shift and suppresses the H_2O_2 burst, finally favoring insect feeding.

Material and methods

Insects and plants

A colony of *B. tabaci* (cryptic species MED) was originally collected from a soybean field in Suzhou, Anhui Province, China. The insects were subsequently maintained on *N. tabacum* cv. K326 plants under controlled laboratory conditions ($25 \pm 1^\circ\text{C}$, 50–70% relative humidity, 16:8 h light:dark photoperiod). For experimental use, *N. benthamiana* and *N. tabacum* plants were grown in a climate-controlled growth chamber ($23 \pm 1^\circ\text{C}$, 16:8 h light: dark cycle).

Phylogenetic analysis of RDP

To identify potential RDP homologs across insect species, we performed a BLASTp search using the BtRDP protein sequence as a query against the predicted proteomes of 30 insect species. Hits were filtered using an E-value cutoff of 10^{-5} . The retrieved RDP homologs were aligned using MAFFT (v7.310) with default parameters [47]. To improve alignment quality, we removed ambiguously aligned regions using Gblocks with default settings. The optimal amino acid substitution model (JTT+G) was determined using ModelTest-NG (v0.1.6). Phylogenetic relationships were inferred using maximum likelihood analysis in RAxML (v0.9.0) with 1000 bootstrap replicates to assess branch support [48]. The final tree was visualized and annotated using iTOL (<https://itol.embl.de/>).

Preparation of anti-BtRDP and anti-NtRLP4 serum

The coding sequence of BtRDP (excluding the signal peptide) was PCR-amplified and cloned into the pET-28a vector (Novagen) to generate a C-terminal 6×His-tagged fusion protein. The recombinant plasmid was transformed into *Escherichia coli* Transetta (DE3) competent cells (TransGen Biotech, Beijing, China). Protein expression was induced with 0.5 mM isopropyl-β-D-thiogalactoside (IPTG) at 28°C for 8 h. Bacterial cells were harvested by centrifugation and lysed via ultrasonication for 30 min. The recombinant BtRDP^{SP}-His protein was purified under native conditions using Ni-NTA agarose (Qiagen, Hilden, Germany) according to the manufacturer's protocol.

Purified BtRDP^{SP}-His protein was used to immunize rabbits (Hua'an Biotechnology, Hangzhou, China) for custom polyclonal antibody production. The anti-NtRLP4 serum, prepared by immunizing rabbits with peptide GQNSDRPISTKNST, was produced via the custom service of Genscript Biotechnology Company (Nanjing, China). The anti-BtRDP serum was subsequently conjugated to Alexa Fluor™ 488 NHS Ester (#A20000, Thermo Fisher Scientific) following standard labeling protocols.

The specificity of anti-BtRDP serum was tested by comparing dsGFP- and dsBtRDP-treated *Bemisia tabaci*. The results showed that dsBtRDP significantly reduced the target gene. Similarly, the specificity of anti-NtRLP4 serum was tested by comparing RNAi-EV and RNAi-NtRLP4 plants, which showed a reduced NtRLP4 expression in RNAi-NtRLP4 plants.

Immunohistochemistry (IHC) staining

Dissected *B. tabaci* heads with attached salivary glands and Parafilm-embedded salivary sheaths were fixed in 4% paraformaldehyde (#E672002, Sango Biotechnology, Shanghai) for 30 min. Salivary sheaths adherent to Parafilm were similarly excised and fixed under identical conditions. After fixation, samples were washed three times with PBST (PBS containing 0.1% Tween-20) and blocked with 10% FBS for 2 h at room temperature. The tissues were then incubated overnight at 4°C with Alexa Fluor 488-conjugated anti-BtRDP serum (1:200 dilution), followed by co-staining with phalloidin-rhodamine (1:500, Thermo Fisher Scientific) for 30 min and 4',6-diamidino-2-phenylindole (DAPI) solution (#ab104139, Abcam, Cambridge, USA) for nuclear visualization. Fluorescent images were acquired using a Leica SP8 confocal microscope (Leica Microsystems) to examine BtRDP localization patterns.

Protein extraction and Western-blotting assays

To identify secreted salivary proteins in *N. tabacum* plants, approximately 200 *B. tabaci* adults were confined in a 3-cm diameter feeding cage and allowed to feed on host plants for 24 h. After feeding, the eggs deposited on the infested tobacco leaves were removed. The leaves showing no visible insect contamination were immediately frozen in liquid nitrogen and ground to a fine powder. The powdered tissue was then homogenized in RIPA Lysis Buffer (#89900, Thermo Fisher Scientific) for protein extraction. To examine the effects of insect feeding on NtRLP expression, groups of approximately 200 *B. tabaci* were allowed to feed on 4-5 true-leaf stage tobacco plants for 12, 24, or 48 h, with uninfested plants serving as controls. For silencing/overexpression efficiency tests, leaf samples were collected at 48 h post-*Agrobacterium* infiltration, along with transgenic tobacco plants at the 4-5 true-leaf stage. For insect tissue analysis, various *B. tabaci* organs were carefully dissected in 1×PBS buffer, including carcasses (20), fat bodies (20), guts (40), salivary glands (60), and ovaries (10), with sample sizes indicated in parentheses. All dissected tissues were transferred to lysis buffer for protein extraction. Total protein concentrations in both insect and plant homogenates were quantified using a BCA Protein Assay Kit (#CW0014S, CwBiotech, Taizhou, China). Equal protein amounts were separated by SDS-PAGE and electro-transferred to PVDF membranes.

Membranes were probed with the following primary antibodies: anti-FLAG (1:10,000, #MA1-91878, Thermo Fisher Scientific), anti-MYC (1:10,000, #MA1-21316, Thermo Fisher Scientific), anti-GFP (1:10,000, #MA5-15256, Thermo Fisher Scientific), anti-NtRLP4 (1:5,000, GenScript Biotechnology, Nanjing, China), and anti-BtRDP serum (1:5,000, Huaan Biotechnology, Hangzhou, China). Following primary antibody incubation, membranes were treated with HRP-conjugated secondary antibodies: goat anti-mouse IgG (1:10,000, #31430, Thermo Fisher Scientific) or goat anti-rabbit IgG (1:10,000, #31460, Thermo Fisher Scientific). Protein bands were visualized using an AI 680 image analyzer (Amersham Pharmacia Biotech, Buckinghamshire, UK), with Ponceau S staining used to verify equal protein loading.

Quantitative real-time PCR (qRT-PCR) analysis

To prepare tissue-specific samples from *B. tabaci*, adult females were dissected to isolate carcasses (10), fat bodies (10), guts (20), salivary glands (40), and ovaries (10). Additionally, different developmental stages, including eggs (50), nymphs (20), pupae (20), and adult males (20) and females (20), were collected for analysis. The number of insects in each sample is indicated in parentheses. To extract RNA from *N. tabacum* plants, the samples were grinded using liquid nitrogen after collection. Total RNA was extracted using the TRIzol Total RNA Isolation Kit (#9109, Takara, Dalian, China) following the manufacturer's instructions. RNA purity and concentration were assessed using a NanoDrop spectrophotometer (Thermo Fisher Scientific). First-strand cDNA was synthesized from equal amounts of RNA (1 µg) using the HiScript II Q RT SuperMix (#R212-01, Vazyme, Nanjing, China). Gene expression analysis was performed using a Roche LightCycler® 480 system (Roche Diagnostics, Mannheim, Germany) with SYBR Green Supermix (#11202ES08, Yeasen, Shanghai, China). The thermal cycling protocol consisted of initial denaturation at 95°C for 5 min, and amplification for 40 cycles of 95°C for 10 s and 60°C for 30 s. Primers were designed with Primer Premier v6.0 (Table S4). *B. tabaci* *Tubulin* and *N. tabacum* *Tubulin* served as internal controls for normalization. Relative gene expression was calculated using the $2^{-\Delta\Delta C_t}$ method [49]. Three to six independent biological replicates, each repeated twice, were performed.

RNA interference

The target gene DNA sequences were amplified using primers listed in Table S4 and subsequently cloned into the pClone007 Vector (#TSV-007, Tsingke Biotechnology, Beijing, China). For dsRNA production, PCR-generated DNA templates containing T7 promoter sequences were transcribed *in vitro* using the T7 High Yield RNA Transcription Kit (#TR101-01, Vazyme Biotech, Nanjing, China). The RNAi experiments were performed according to established methods [50]. Briefly, newly emerged adult whiteflies were anesthetized with carbon dioxide for 5-10 seconds. Approximately 50 nL of dsRNA solution (2 µg/µL) was microinjected into the mesothorax of each insect using a

FemtoJet microinjection system (Eppendorf, Hamburg, Germany). Following injection, the whiteflies were maintained on healthy *N. tabacum* leaves under standard rearing conditions. Only actively moving insects were selected for subsequent analyses. Gene silencing efficiency was evaluated on day 4 post-injection using both qRT-PCR and Western-blotting approaches.

Generation of transgenic plants

We generated two transgenic *N. tabacum* lines expressing different BtRDP variants: one containing the full coding sequence (oeBtRDP) and another lacking the signal peptide (oeBtRDP^{SP}), which may affect protein secretion. The target sequences, along with the full-length NtRPL4 coding sequence and an NtRPL4 hairpin RNAi construct, were each cloned into the pBWA(V)HS vector. The oeBtRDP and oeBtRDP^{SP} were fused with C-terminal flag tags, while no tag was fused to oeNtRPL4. All constructs were transformed into *A. tumefaciens* strain GV3101 and used to generate transgenic tobacco plants (K326 background) through *Agrobacterium*-mediated transformation performed by Wuhan Boyuan Biological Company. Transgenic lines were verified by qRT-PCR and western blotting, with empty vector-transformed plants serving as controls, and two independent lines for each construct were selected for further experiments.

Target gene silencing in plants

The hairpin RNAi construct targeting *NtSOBIR1* was generated by cloning the specific fragment into the pBWA(V)HS plasmid using BsaI and Eco31I restriction enzymes. The recombinant vector was then transformed into *A. tumefaciens* strain GV3101 and subsequently infiltrated into *N. tabacum* leaves using the agroinfiltration method. Following a 48-h incubation period under standard growth conditions, silencing efficiency was quantitatively evaluated by qRT-PCR analysis.

Transcriptomic sequencing

To investigate the transcriptional effects of NtRPL4, BtRDP, and BtRDP^{SP} in *N. tabacum*, leaf samples from EV, oeRPL4#1, oeBtRDP#1, and oeBtRDP^{SP}#1 transgenic plants were homogenized in TRIzol Reagent (#10296018, Invitrogen, Carlsbad, CA, USA) for total RNA extraction following the manufacturer's protocol. Afterwards, total RNA was extracted according to the instructions of the manufacturer, and the RNA samples were sent to Novogene Institute (Novogene, Beijing, China) for transcriptomic sequencing. Briefly, poly(A)⁺ RNA was isolated from 20 µg pooled total RNA using oligo(dT) magnetic beads, followed by fragmentation in divalent cation buffer at 94°C for 5 min. First-strand cDNA synthesis was performed using N6 random primers, with subsequent second-strand synthesis to generate double-stranded cDNA. After end-repair and Illumina adapter ligation, the products were PCR-amplified (15 cycles) and purified using a QIAquick PCR Purification Kit (#28104, Qiagen, Hilden, Germany). The resulting cDNA libraries were quantified and quality-checked before paired-end sequencing on an Illumina NovaSeq 6000 platform. All raw sequencing data were deposited in the National Genomics Data Center under accession number PRJCA025907.

Analysis of transcriptomic data

The raw sequencing reads were quality-filtered using Illumina's internal software, and the resulting clean reads from each cDNA library were aligned to the *N. tabacum* reference genome in Sol Genomics Network (https://solgenomics.net/ftp/genomes/Nicotiana_tabacum/edwards_et_al_2017/) using HISAT v2.1.0 [51]. Low-quality alignments were removed using SAMtools (v1.7) with default parameters [52]. Transcript abundance was quantified as Transcripts Per Million (TPM) using Cufflinks (v2.2.1) [53]. Differential expression analysis was performed with DESeq2 (v2.2.1) [54], identifying significantly DEGs using thresholds of $|\log_2(\text{fold change})| > 1$ and adjusted p -value < 0.05 . PCA of transcriptome profiles was conducted using a custom R script plotPCA (https://github.com/franco-ye/TestRepository/blob/main/PCA_by_deseq2.R). For functional annotation, KEGG pathway enrichment analysis was performed in TBtools (v2.083) using a one-sided hypergeometric test:

$$P = 1 - \sum_{i=0}^{m-1} \left(\frac{\binom{M}{i} \binom{N-M}{n-i}}{\binom{N}{n}} \right), [55].$$
 In this software, enriched P -values were calculated according to one-

sided hypergeometric test: with N representing the number of gene with KEGG annotation, n standing for the number of DEGs in N , M indicating the number of genes in each KEGG term, and m suggesting the number of DEGs in each KEGG term.

Insect bioassays

To survivorship analysis, newly emerged *B. tabaci* adults were microinjected with dsRNA and placed on *N. tabacum* leaves. After 24 h, any dead individuals (potentially due to injection injury) were removed. For each treatment, 20-30 surviving adults were transferred to leaf cages, and mortality was recorded daily for 10 days. Three independent biological replications were performed.

For fecundity analysis, dsRNA-treated virgin females were paired with untreated males in leaf cages. Five such pairs were maintained per cage ($n=10-12$ cages per treatment). Egg deposition was quantified daily for 3 consecutive days under controlled conditions. Each male-female pair was considered an experimental replicate.

For host preference test, detached *N. tabacum* leaves with moist cotton-wrapped petioles were arranged in a 50-cm diameter choice arena with a central release chamber. Forty adult whiteflies were released per chamber, and settling preference was recorded at 3, 6, 12, 24, 36, and 48 h post-release. The experiment included 10-12 replicates per treatment, with leaf positions randomized between replicates to eliminate positional bias.

EPG recording and analysis

Feeding behavior of *B. tabaci* was monitored using a GiGA-8d EPG amplifier (Wageningen Agricultural University; 10 T Ω input resistance, <1 pA input bias current) with 50 $\times$ gain and ± 5 V output range. Prior to recording, dsRNA-treated whiteflies were starved for 12 h on moist filter paper, briefly anesthetized with CO₂ (10 s), and connected to the amplifier via a gold wire electrode (20 μ m diameter $\times$ 5 cm length) attached to the pronotum using conductive silver glue. Plant electrodes consisted of copper wires (2 mm diameter $\times$ 10 cm length) inserted into the soil of potted *N. tabacum*. Recordings were conducted for 8 h in a Faraday cage (120 $\times$ 75 $\times$ 67 cm; Dianjiang, Shanghai), with only insects surviving the full duration retained for analysis.

Recorded waveforms were processed using PROBE 3.4 software (Wageningen Agricultural University) and categorized into four distinct feeding phases: nonpenetration (np), pathway duration (C), phloem salivation (E1), and phloem ingestion (E2) as defined in a previous study [31]. Each treatment included ≥ 10 biological replicates, with dsGFP-injected and untreated whiteflies serving as controls.

Agrobacterium-mediated plant transformation in *N. tabacum* and *N. benthamiana*

The recombinant expression vectors were introduced into *A. tumefaciens* strain GV3101 via heat shock transformation. Transformed cells were selected on LB agar plates supplemented with 50 μ g/mL kanamycin and 10 μ g/mL rifampicin, followed by incubation at 28°C for 60 h. Positive colonies were inoculated into liquid LB medium with the same antibiotics and grown to log phase. Bacterial cells were then pelleted by centrifugation at 2,400 g for 2 min and resuspended in induction buffer (10 mM MgCl₂, 10 mM MES [pH 5.6], 200 μ M acetosyringone) adjusted to specific OD₆₀₀ values depending on the experiment.

For standard experiments, bacterial suspensions were normalized to OD₆₀₀ = 1.0, while degradation assays used three concentrations (OD₆₀₀ = 0.05, 0.3, and 1.0). Equal volumes of selected bacterial suspensions were mixed and infiltrated into leaves of 4–5-week-old *N. tabacum* and *N. benthamiana* plants using a needleless syringe. Infiltrated plants were maintained under controlled conditions for subsequent analysis.

Diaminobenzidine (DAB) staining

To detect H₂O₂ accumulation in *N. tabacum* leaves, leaf samples were immersed in 1 mg/mL 3,3'-diaminobenzidine tetrahydrochloride (DAB-HCl, #D8001, Sigma-Aldrich, St. Louis, MO, USA) solution (pH 3.8) for 6 hours at room temperature, allowing H₂O₂-dependent brown polymerization to occur. Following incubation, the DAB solution was replaced with 100% ethanol for chlorophyll removal, and samples were decolorized overnight at 65°C. The destained leaves were then photographed under standardized conditions using a Canon EOS 80D digital camera (Canon Inc., Tokyo, Japan).

Interaction assays between two proteins

In the Y2H screening assay, the coding sequence of BtRDP^{SP} was cloned into the pGBKT7 bait vector (Clontech, USA), while a *N. benthamiana* cDNA library was constructed in the pGADT7 prey vector (Biogene Biotech, Shanghai, China). These recombinant vectors were co-transformed into yeast strain Y2HGold and plated on QDO medium (SD/-Ade/-His/-Leu/-Trp, #630428, Takara) for 3 days at 30°C. Positive colonies were cultured in QDO liquid medium, and plasmids were extracted using a yeast DNA kit (#DP112-02, TIANGEN, Beijing, China) before transformation into *E. coli* DH5α (#TSC-C14, Tsingke) for sequencing (YouKang Biotech, China) to identify interacting partners.

In the Y2H point-to-point verification assay, BtRDP^{SP} and NISP104^{SP}, and mutant variants of NtRPL4/SIRLP4/ OsRPL4 were cloned into either pGBKT7 or pGADT7 vectors using primers from Table S4. Yeast co-transformants were first selected on DDO medium (SD/-Leu/-Trp) (#630417, Takara) for 3 days at 30°C, then spotted onto QDO medium to confirm interactions through growth after another 3 days of incubation.

In the Co-IP assays, Full-length sequences of BtRDP, NISP104, NtSOBIR1, NtRPL4, SIRLP4, and OsRPL4 were cloned into lic-myc or lic-flag vectors for C-terminal tagging, with BtFTSP-flag serving as a negative control. Five-week-old *N. benthamiana* leaves were agroinfiltrated with these constructs or collected from oeBtRDP transgenic plants. Total proteins were extracted from 1 g frozen tissue in IP lysis buffer (#87788, Thermo Scientific) with protease inhibitors (#56079200, Roche, Switzerland), followed by centrifugation at 1,000 × *g* for 20 min. Cleared lysates were incubated with anti-flag beads (#L00432-1, GenScript, Nanjing, China) for 4 h at 4°C, washed with PBS, and eluted in 2× SDS-PAGE buffer (500 mM Tris-HCl, pH=6.8, 50% glycerol, 10% SDS, 1% bromophenol blue, and 2% β-mercaptoethanol) for Western-blotting assay.

In the BiFC assay, BtRDP^{SP}, BtFTSP^{SP}, NtRPL4, and NtCf9-like were cloned into the pCV-cYFP or pCV-nYFP vectors, respectively. After transforming *A. tumefaciens* GV3101, pairwise combinations were co-infiltrated into *N. benthamiana* leaves. Following 36–48 h incubation in a growth chamber, reconstituted YFP fluorescence was visualized using a Leica SP8 confocal microscope with standard filter sets for YFP detection.

RPL degradation by salivary effectors among different herbivorous insects

To examine salivary protein effects on RLPs, we co-infiltrated *A. tumefaciens* GV3101 strains carrying various constructs into *N. benthamiana* leaves. The recombinant vectors expressed C-terminally tagged proteins: BtRDP-flag, NISP104-flag, GUS-flag, NISP7-flag, and myc-tagged RLPs (NtRPL4, SIRLP4, OsRPL4, SlCf9, NtCf9). Bacterial suspensions were mixed at predetermined OD₆₀₀ ratios before infiltration, with leaf samples collected 48 hours post-infiltration for subsequent analysis.

For confocal microscopy studies, *N. benthamiana* leaves were co-infiltrated with *A. tumefaciens* carrying: (1) NtRPL4-GFP plus BtRDP-mCherry, (2) NtRPL4-GFP plus RFP-mCherry, or (3) NtRPL4-GFP plus BtFTSP-mCherry. The samples were imaged by confocal microscopy at 48 h post injection. The same parameters were used in photographing.

This study also conducted a treatment on *N. benthamiana* plants that were overexpressing NtRPL4-myc using purified proteins. Briefly, the BtRDP^{SP} and GFP coding sequences were cloned into pET-28a for 6×His-tagged expression in *E. coli* Transetta (TransGen Biotech). Protein expression was induced with 0.5 mM IPTG at 25°C for 8 h, followed by purification using Ni-NTA agarose and buffer exchange into PBS using a 3-kDa molecular-weight cutoff Amicon Ultra-4 Centrifugal Filter Device (Millipore, MA, USA). Different concentrations of purified proteins were infiltrated into leaves of *N. benthamiana* plants stably expressing NtRPL4-myc, with samples collected 24 h post-treatment.

To investigate degradation pathways, we infiltrated *N. benthamiana* leaves with: 50 μM MG132 (#M7449, Sigma-Aldrich, Steinheim, Germany), 5 μM 3-MA (#M9281, Sigma-Aldrich), or 10 nM BAF (#BML-CM110-0100, Enzo Life Sciences, Farmingdale, USA) at 24 h post-agroinfiltration. The samples were incubated for additional 24 h. Protein levels were detected using Western-blotting assay.

***In planta* ubiquitination assay**

Ubiquitination detection in *N. benthamiana* was performed according to the methods previously reported [56]. The recombinant vector for transient expression of the HA-fused ubiquitin protein was kindly provided by Dr. Chao Zheng (Ningbo University). Myc-tagged proteins were co-expressed with HA-UBQ by agroinfiltration, and 50 μM MG132 was injected into the *N. benthamiana* leaves one day post *A. tumefaciens* infiltration. Immunoprecipitation was performed using anti-myc beads (ytma-20, ChromoTek, Planegg-Martinsried, Germany). Ubiquitination signals were detected by immunoblot with anti-HA antibody (HT301, TransGen Biotech, Beijing, China).

Scanning electron microscopy (SEM)

B. tabaci adults were allowed to feed on artificial diet solutions for 24 h. Parafilm sections containing salivary sheaths were carefully excised and rinsed with 1× PBS (pH 7.4) to remove residual diet. The samples were mounted on aluminum stubs, desiccated under vacuum, and sputter-coated with gold before imaging with a TM4000 II Plus SEM (Hitachi, Tokyo, Japan). Salivary sheath length (from apex to base) was measured, with 20 sheaths analyzed per treatment to ensure statistical reliability.

Statistical analysis

The log-rank test (SPSS Statistics 19, Chicago, IL, USA) was used to determine the statistical significance of survival distributions. Two-tailed unpaired Student's *t*-test (comparisons between two groups) or one-way ANOVA test followed by Tukey's multiple comparisons test (comparisons among three groups) was used to analyze the results of qRT-PCR, fecundity analysis, and host choice analysis. The EPG data were first checked for the normality and homogeneity of variance, and those not fitting a normal distribution were log₁₀ transformed before Student's *t* test, as described in previous studies [31]. Statistical tests, including Student's *t*-test and one-way ANOVA, and data visualization were conducted in GraphPad Prism 9.

Supplementary Figures

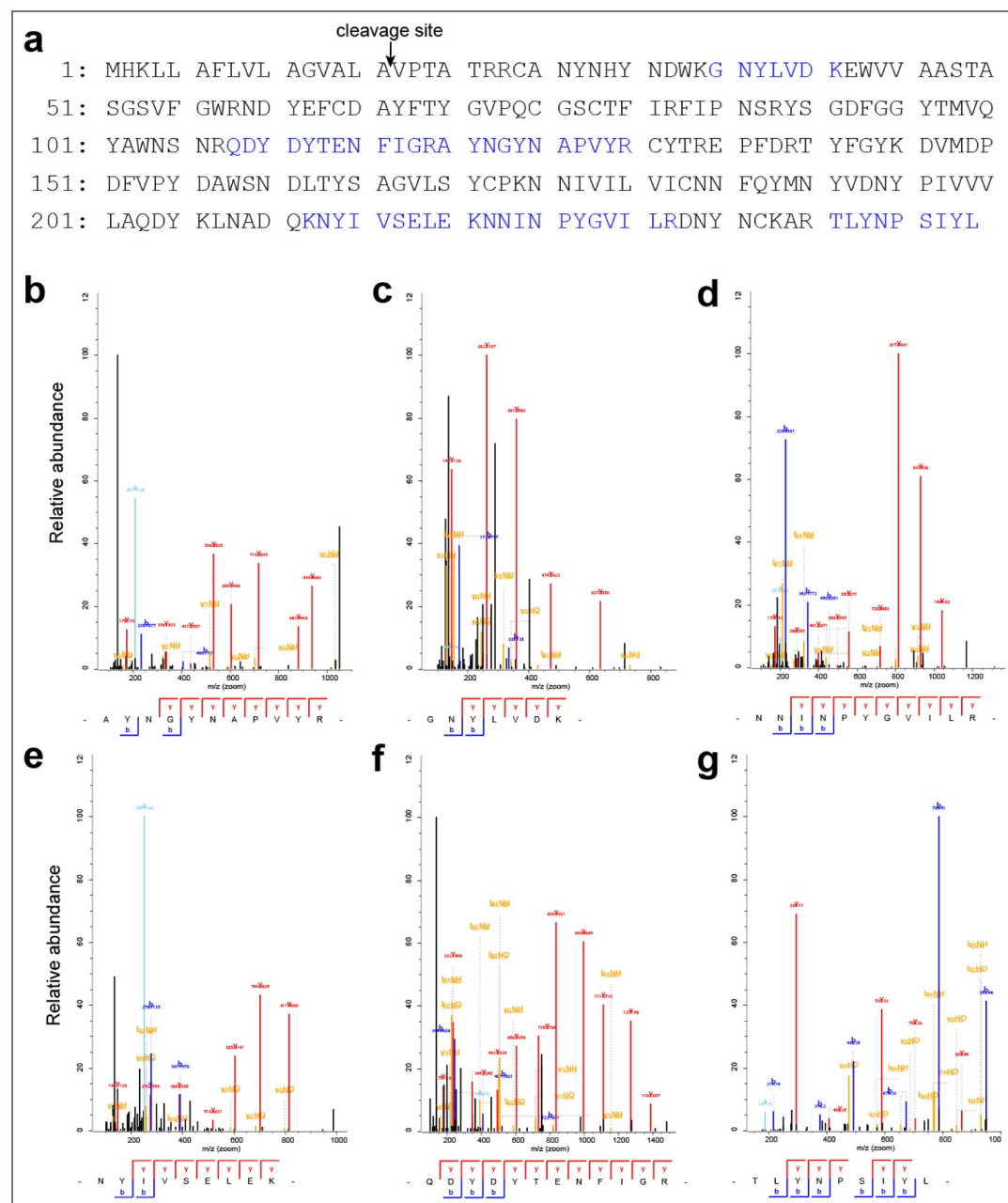


Fig. S1. Characteristic of BtRDP. (a) Deduced amino acid sequence of BtRDP. Arrow indicates the signal peptide cleavage site. **(b-g)** Mass spectrums of the identified unique peptides in watery saliva of *Bemisia tabaci*.

Fig. S2. Analysis of insect RDP.

(a) Phylogenetic tree of RDP sequences from Aleyrodidae species. The trees are constructed by RAxML v0.9.0 using the maximum likelihood method with 1000 bootstrap replicates. Nodes with bootstrap values greater than 50 are displayed. (b) Sequence alignments of RDPs. The amino acid sequence of RDPs used for phylogenetic tree construction are aligned using ClusterX software. Black shades indicate the conserved regions.

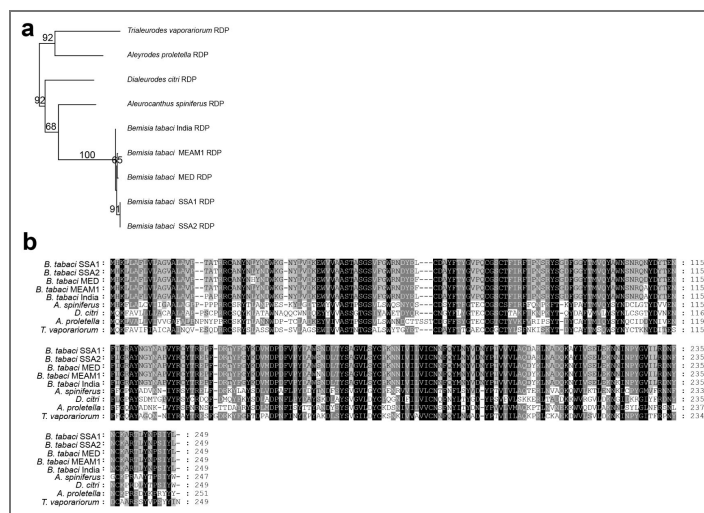


Fig. S3. Expression patterns of *Bemisia tabaci* BtRDP.

Transcripts per million (TPM) expression values of *BtRDP* in different tissues (a) and at different developmental stages (b) are determined based on the transcriptomic data. Data are presented as mean values $\pm$ SEM (n=3 independent biological replicates).

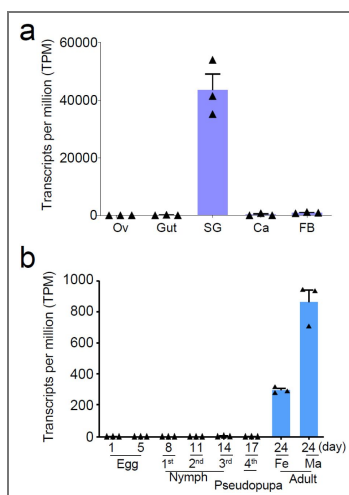


Fig. S4. Expression patterns of *BtActin* and *Bt18s rRNA*.

The transcript level of each gene in different tissues (a) and at different developmental stages (b) is quantified by qRT-PCR. *Bemisia tabaci tubulin* is used as an internal control. The relative quantitative method ($2^{-\Delta\Delta C_t}$) is used to evaluate the quantitative variation. Ov, ovary; SG, salivary gland; Ca, carcass; FB, fat body; Ny, nymph; Ps, pseudopupa; Fe, female; Ma, male. Data are presented as mean values $\pm$ SEM (n=3 independent biological replicates). Different lowercase letters indicate statistically significant differences at $P < 0.05$ level according to one-way ANOVA test followed by Tukey's multiple comparisons test.

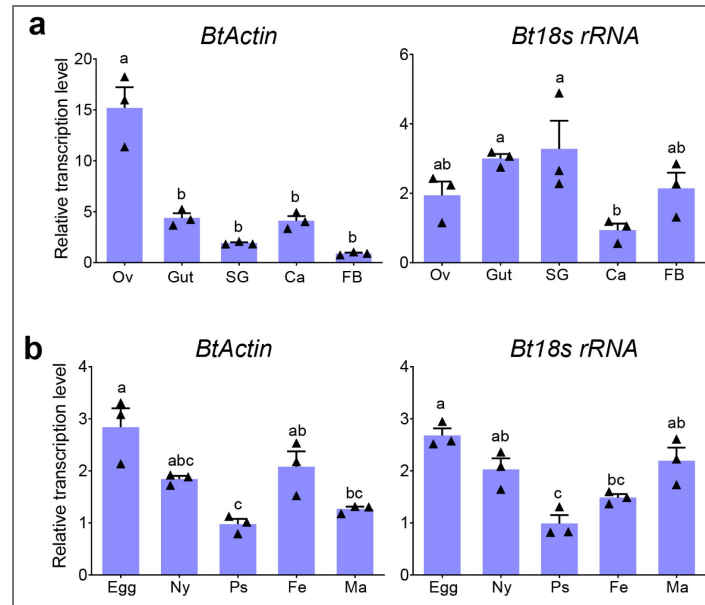


Fig. S5. BtRDP is efficiently and specifically suppressed by dsBtRDP.

(a) Relative transcript level of *BtRDP*, *BtFTSP*, *BtE3*, and *Bt56* after dsRNA treatments. *Bemisia tabaci* are treated with dsGFP and dsBtRDP. The expression pattern of each gene is determined 4 days post treatment using qRT-PCR methods. *B. tabaci tubulin* is used as an internal control. The relative quantitative method ($2^{-\Delta\Delta C_t}$) is used to evaluate the quantitative variation. Data are presented as mean values $\pm$ SEM (n= 3 independent biological replicates) (b) Protein level of BtRDP after dsRNA treatments. Actin is used as an internal control. Two independent biological replicates are displayed. (c) Immunohistochemical staining of BtRDP after dsRNA treatment. The salivary gland (SG) and its nearby tissues are collected from dsGFP- and dsBtRDP-treated *Bemisia tabaci*. The samples are incubated with anti-BtRDP serum conjugated with Alexa Fluor™ 488 NHS Ester (green) and actin dye phalloidinrhodamine (red), and examined by Leica SP8. The nucleus is stained with DAPI (blue). The lower image in each treatment represents the enlarged images of the boxed area in the upper image. Experiments are repeated twice, with each inspecting more than 10 salivary glands.

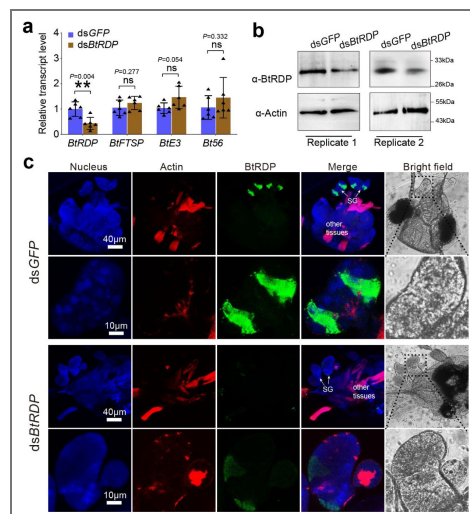


Fig. S6. Effects of dsRNA treatment on insect survivorship and salivary sheath formation.

(a, b) Effects of dsRNA treatment on salivary sheath formation. Newly emerged *Bemisia tabaci* adults are injected with dsGFP and dsBtRDP. The dsRNA-treated *B. tabaci* are fed on artificial diets. The salivary sheaths left on paraform are inspected by scanning electron microscopy (a), and the length of salivary sheath is measured from the top to base of salivary sheath (b). Data are presented as mean values $\pm$ SEM. Twenty salivary sheaths from each treatment are measured. *P*-value is determined by two-tailed unpaired Student's *t* test. **P* < 0.05. (c) Effects of dsRNA treatment on insect survivorship. A group of 20-30 *B. tabaci* are placed in a leaf cage, and their mortality is recorded for ten consecutive days. Three independent biological replications are performed. Differences in survivorship between the two treatments are tested by log-rank test. ns, not significant.

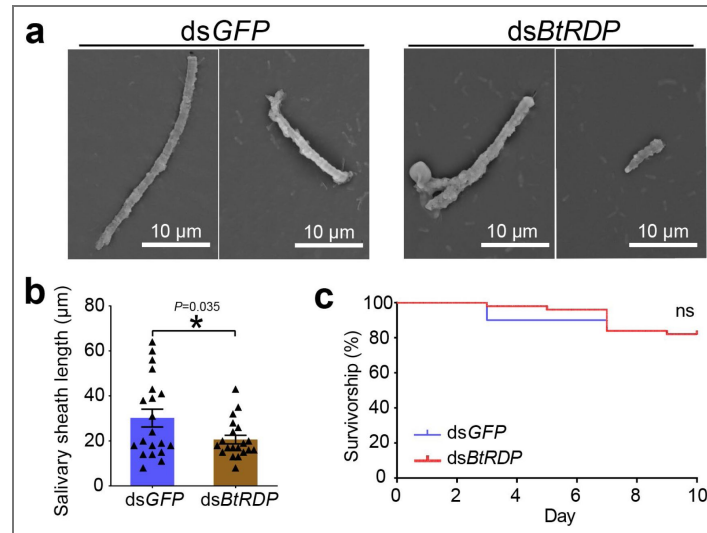


Fig. S7. Typical EPG waveforms for *Bemisia tabaci* feeding on *Nicotiana tabacum*.

The *B. tabaci* feeding behavior can be classified into nonpenetration (np), pathway duration (C), phloem salivation (E1) and phloem ingestion (E2).

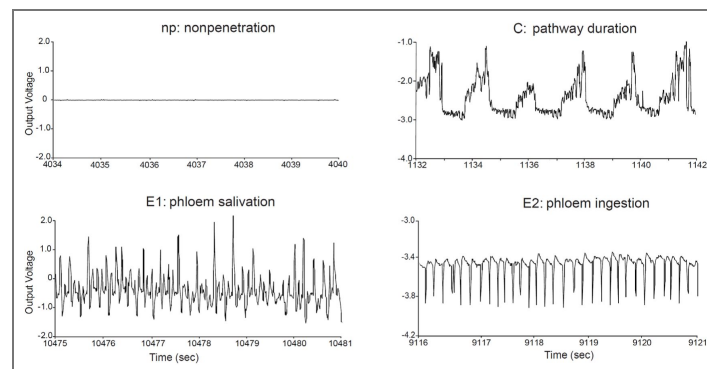


Fig. S8. Effects of empty vector (EV) transgenic plants on *Bemisia tabaci*.

Two-choice experiment is performed to investigate the attraction of wild type (WT) and EV transgenic plants to *B. tabaci*. A group of 40 female *B. tabaci* are released into a device containing EV and WT leaves. The number of insects settling on each leaf is counted at 3, 6, 12, 24, 36, and 48 h (a). After 48 h, the number of eggs on each leaf is counted (b). Data are presented as mean values $\pm$ SEM. Ten independent biological replicates are performed. *P*-values are determined by two-tailed unpaired Student's *t* test. ns, not significant.

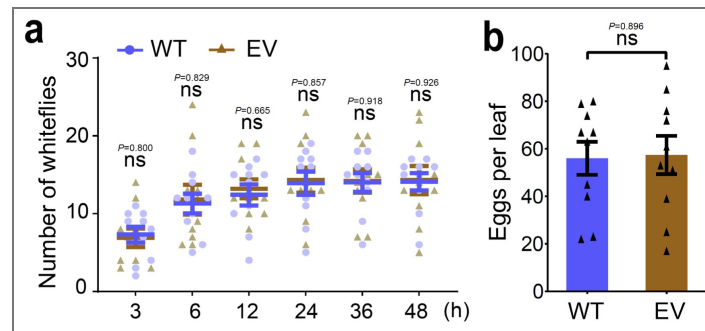


Fig. S9. Effects of BtRDP and BtRDP^{SP} overexpression on *Bemisia tabaci*.

Insect performance on oeBtRDP#2 (a, b) and oeBtRDP^{SP}#2 (c, d) transgenic plants are tested. (a, c) Comparison of insect reproduction on empty vector (EV) and oeBtRDP#2/oeBtRDP^{SP}#2 transgenic plants. Five *B. tabaci* individuals are confined to indicated plants for 3 days, and the oviposited eggs are counted. Twelve independent biological replicates are performed. (b, d) Attraction of EV and oeBtRDP#2/oeBtRDP^{SP}#2 leaves to *B. tabaci* in a two-choice equipment. A group of 40 female *B. tabaci* are released into a device containing indicated leaves. The number of insects settling on each leaf is counted at 3, 6, 12, 24, 36, and 48 h. After 48 h, the number of eggs on each leaf is counted. Data are presented as mean values $\pm$ SEM. Ten independent biological replicates are performed. *P*-values are determined by two-tailed unpaired Student's *t* test. ****P* < 0.001.

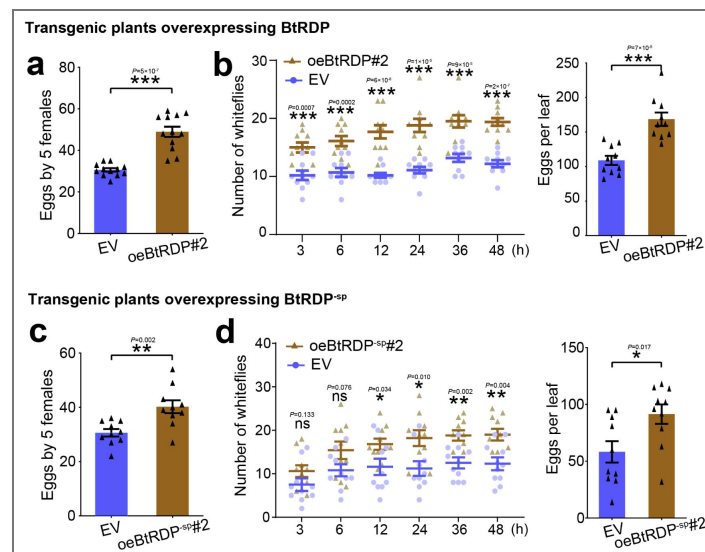


Fig. S10. Expression patterns of *NtRPL4* in response to *Bemisia tabaci* infestation.

Relative transcript level of *NtRPL4* in response to *B. tabaci* infestation are determined by qRT-PCR. Data are presented as mean values $\pm$ SEM. Different lowercase letters indicate statistically significant differences at $P < 0.05$ level according to one-way ANOVA test followed by Tukey's multiple comparisons test. Three independent biological replicates are performed.

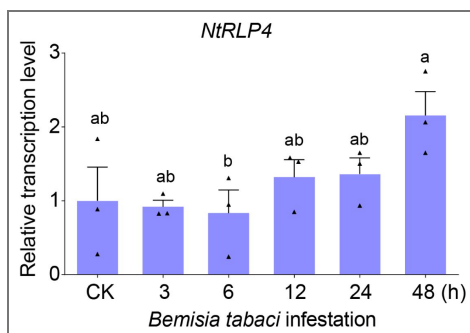


Fig. S11. Sequence alignments of SICf9 and NtCf9.

The amino acid sequence of SICf9 and NtCf9 are aligned using ClusterX software. Black shades indicate the conserved regions.

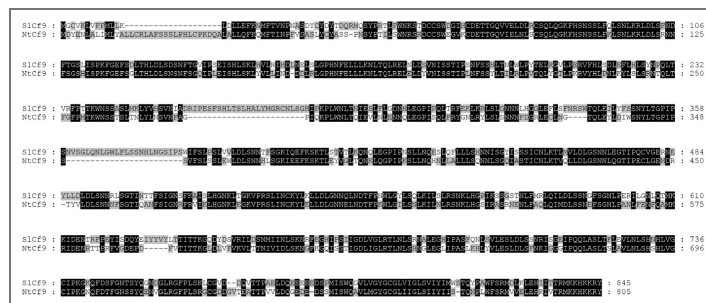


Fig. S12. BtRDP interacts with NtRLP4.

(a) Yeast two hybrid assays showing the interaction between NtRLP4 and different whitefly salivary proteins (BtFTSP^{SP}, BtSP16.3^{SP} and BtSP37.4^{SP}, without signal peptides), as well as BtRDP^{SP} and truncated NtRLP4. The different combinations of constructs are transformed into yeast cells, and are grown on the selective medium SD/-Trp/-Leu (DDO), and the interactions are tested with SD/-Trp/-Leu/-His/-Ade (QDO). (b) Expression of three control salivary proteins at the protein level. The target bands are indicated by asterisks. (c) Bimolecular fluorescence complementation assay showing the specific interaction between BtRDP and NtRLP4. YFP fluorescence is observed when co-expressing the N-terminal nYFP tag fused NtRLP4 and N-terminal cYFP tag fused BtRDP^{SP}. Bar =40 μ m. (d) Co-immunoprecipitation (Co-IP) assays on oeBtRDP transgenic plants. oeBtRDP transgenic plants that overexpressing BtRDP-flag are incubated with anti-flag beads. Endogenous NtRLP4 can be immunoprecipitated by BtRDP-flag in oeBtRDP transgenic plants, while the flag tag alone in EV plants fails to immunoprecipitate endogenous NtRLP4. Rubisco staining (Rbcl) is used to visualize the amount of sample loading.

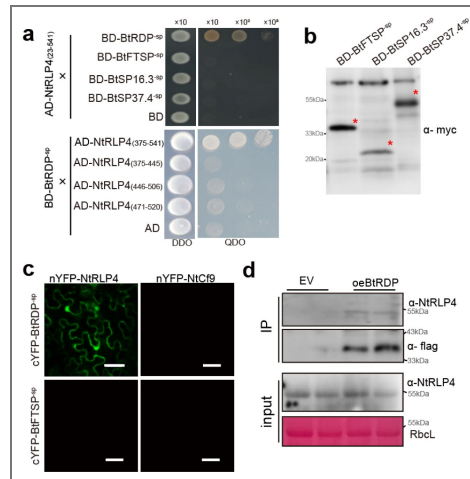
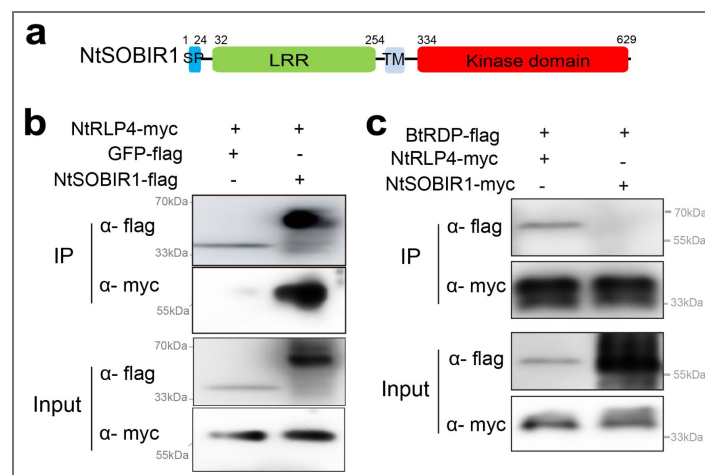


Fig. S13. NtSOBIR1 interacts with NtRLP4 but not BtRDP.

(a) Domain organization of NtSOBIR1. NtSOBIR1 contains a predicted N-terminal signal peptide (SP), a leucine-rich repeat (LRR) domain, a transmembrane (TM) domain, and a kinase domain. (b) Co-immunoprecipitation assay showing the interaction between NtSOBIR1 and NtRLP4. (c) Co-immunoprecipitation assay showing the interaction between NtRLP4 and BtRDP, but not NtSOBIR1 and BtRDP. The complete coding region of *NtSOBIR1* and *NtRLP4* are fused with flag and myc tags at the C-terminal ends. Total proteins are extracted from *N. benthamiana* leaves co-expressing NtRLP4-myc with GFP-flag or NtSOBIR1-flag. Precipitation is performed using flag beads. The samples are probed with anti-flag and anti-myc antibodies for immunoblot analysis. Experiments are repeated twice with the similar results.



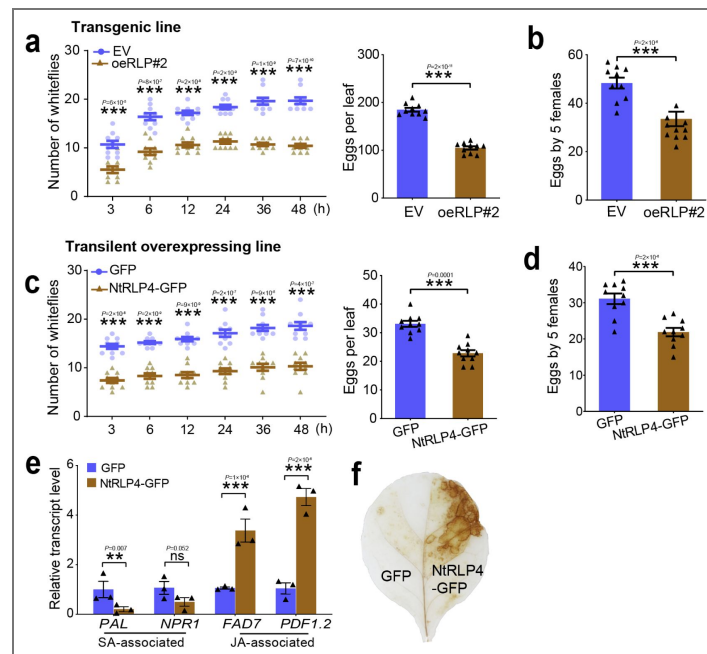


Fig. S14. Effect of NtRPL4 overexpression on *Bemisia tabaci* performance.

(a) Attraction of empty vector (EV) and oeNtRPL4#2 (oeRPL#2) transgenic *Nicotiana tabacum* to *B. tabaci* in a two-choice equipment. A group of 40 female *B. tabaci* are released into a device containing oeRPL#2 and EV leaves. The number of insects settling on each leaf is counted at 3, 6, 12, 24, 36, and 48 h. After 48 h, the number of eggs on each leaf is counted. **(b)** Comparison of insect reproduction on EV and oeRPL#2 transgenic plants. Five *B. tabaci* individuals are confined to indicated plants for 3 days, and the oviposited eggs are counted. **(c)** Attraction of leaves transiently overexpressing GFP and NtRPL4-GFP to *B. tabaci* in a two-choice equipment. **(d)** Comparison of insect reproduction on *N. tabacum* leaves transiently overexpressing GFP and NtRPL4-GFP. **(e)** Relative transcript level of salicylic acid (SA)- and jasmonic acid (JA)-associated genes in *N. tabacum* leaves transiently overexpressing GFP and NtRPL4-GFP. PAL, phenylalanine ammonia lyase; NPR1, nonexpressor of pathogenesis-related protein 1; FAD7, fatty acid desaturase 7; PDF1.2, plant defensin 1.2. **(f)** Transiently overexpressing NtRPL4-GFP induces H₂O₂ accumulation. For bioassays in **(a)**, **(b)**, **(c)**, and **(d)**, ten independent biological replicates are performed. For qPCR analysis in **(e)**, three independent biological replicates are performed. Data are presented as mean values $\pm$ SEM. *P*-values are determined by two-tailed unpaired Student's *t* test. ****P* < 0.001. The experiment in **(f)** is repeated five times with the similar results.

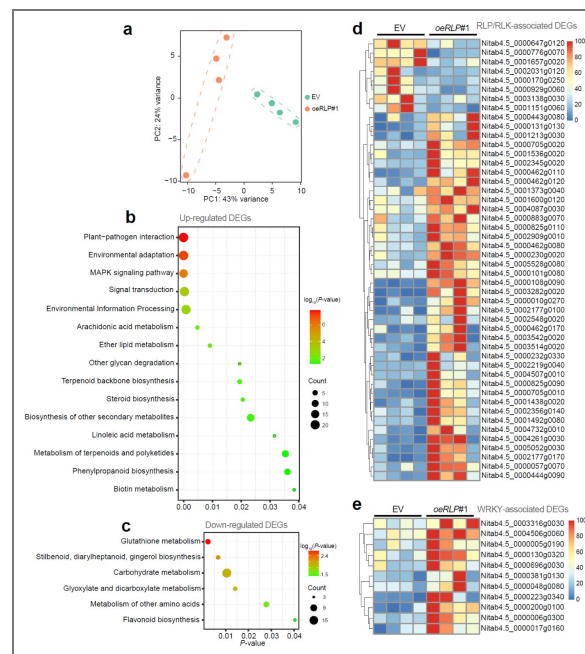


Fig. S15. Transcriptomic comparison of empty vector (EV) and oeNtRLP4#1 (oeRLP#1) transgenic plants.

(a) Principal component analysis (PCA) of gene expression patterns in EV and oeRLP#1 transgenic *Nicotiana tabacum* plants. The first two principal components (PC1 and PC2) based on transcriptomic results are shown. Four independent biological replicates are performed. (b, c) Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway enrichment analysis of differentially expressed genes (DEGs) that significantly up-regulated (b) and down-regulated (c). Enriched *P*-values are calculated according to one-sided hypergeometric test using TBtools software. (d, e) Expression patterns of DEGs that annotated as RLK/RLP (d) or WRKY transcription factor (e). Gene expression patterns are illustrated by a heatmap. The max transcripts per million value of each gene is set as 100.

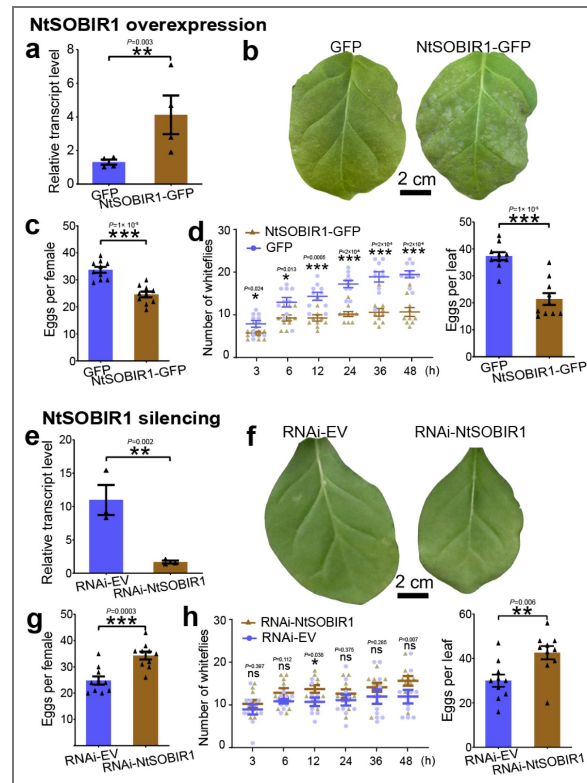


Fig. S16. Effect of NtSOBIR1 overexpression and silencing on *Bemisia tabaci* performance.

(a) Validation of NtSOBIR1 transient overexpression using qRT-PCR ($n=4$ biological replicates). (b) NtSOBIR1 transient overexpression induces cell death phenotype five days post agro-injection. (c) Comparison of insect reproduction on *Nicotiana tabacum* transiently overexpressing GFP and NtSOBIR1-GFP. Five *B. tabaci* individuals are confined to indicated plants for 3 days, and the oviposited eggs are counted. (d) Attraction of GFP and NtSOBIR1-GFP leaves to *B. tabaci* in a two-choice equipment. A group of 40 female *B. tabaci* are released into a device containing GFP and NtSOBIR1-GFP leaves. The number of insects settling on each leaf is counted at 3, 6, 12, 24, 36, and 48 h (left). After 48 h, the number of eggs on each leaf is counted (right). (e) Silencing efficiency of hairpin NtSOBIR1 using qRT-PCR ($n=3$ independent biological replicates). (f) Transiently silencing NtSOBIR1 affect growth of infiltrated leaves. (g) Comparison of insect reproduction on *Nicotiana tabacum* leaves infiltrated with hairpin empty vector (EV) and NtSOBIR1. (h) Attraction of hairpin EV and NtSOBIR1-treated leaves to *B. tabaci* in a two-choice equipment. The experiments in (b) and (f) are repeated ten times with the similar results. In (c), (d), (g), and (h), ten independent biological replicates are performed. P -values are determined by two-tailed unpaired Student's t test. $***P < 0.001$; $**P < 0.01$.

Fig. S17. Effects of RLP silencing on *Bemisia tabaci* performance.

(a) Detection of NtRLP4 protein in empty vector (EV) and NtRLP4-silenced (RNAi-RLP) transgenic *Nicotiana tabacum*. Rubisco staining (Rbcl) is conducted to visualize the amount of sample loading. (b, c) Attraction of EV and RNAi-RLP plants to *B. tabaci* in a two-choice equipment. Two NtRLP4-silenced plants (RNAi-RLP#1 in b; RNAi-RLP#2 in c) are assayed. A group of 40 female *B. tabaci* are released into a device containing two leaves. The number of insects settling on each leaf is counted at 3, 6, 12, 24, 36, and 48 h. After 48 h, the number of eggs on each leaf is counted. Ten independent biological replicates are performed. Data are presented as mean values $\pm$ SEM. *P*-values are determined by two-tailed unpaired Student's *t* test. ****P* < 0.001.

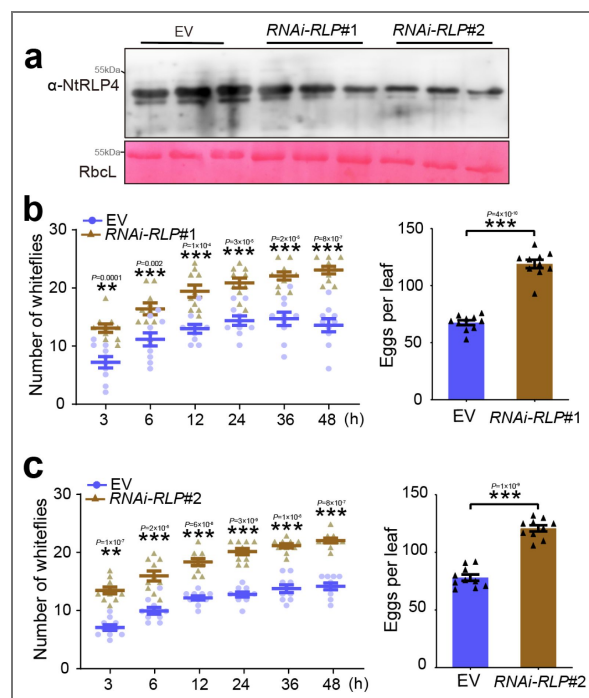


Fig. S18. Influence of BtRDP on NtRLP4 accumulation by fluorescent analysis.

The NtRLP4-GFP is transiently co-expressed with red fluorescent protein (RFP)-mCherry, BtFTSP-mCherry, or BtRDP-mCherry via agroinfiltration. The samples are imaged by confocal microscopy at 48 h post injection. RFP-mCherry and BtFTSP-mCherry are used as negative controls. Three independent biological replicates are performed, and three representative images are taken in each biological replicate. Fluorescence intensity is measured using ImageJ. The intensity values from three biological replicates are calculated and the mean value in the upper left image is set at 1.0. Representative fluorescence images are displayed. The small triangle indicates the different concentrations ($OD_{600} = 0.1$ and 1.0) of *Agrobacterium*. Bar = 40 μ m.

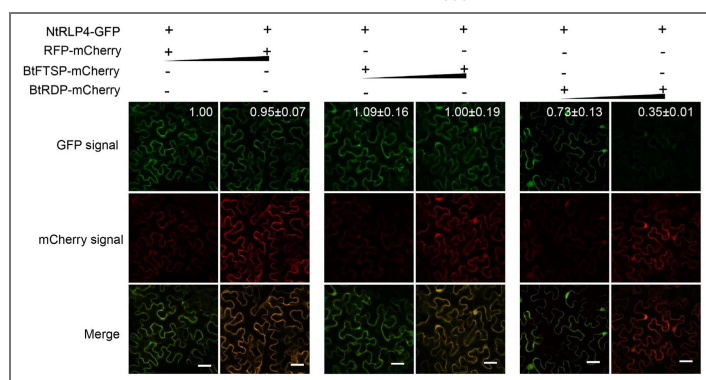


Fig. S19. Effects of BtRDP on suppressing NtRNP4-associated plant defenses.

(a) Comparison of insect reproduction on *Nicotiana tabacum* plants transiently overexpressing GFP along and BtRDP-mCherry/NtRNP4-GFP. Five *Bemisia tabaci* individuals are confined to indicated plants for 3 days, and the oviposited eggs are counted. (b) Attraction of GFP- and BtRDP-mCherry/NtRNP4-GFP-expressed leaves to *B. tabaci* in a two-choice equipment. A group of 40 female *B. tabaci* are released into a device containing two leaves. The number of insects settling on each leaf is counted at 3, 6, 12, 24, 36, and 48 h. After 48 h, the number of eggs on each leaf is counted. Data are presented as mean values $\pm$ SEM. Ten independent biological replicates are performed. *P*-values are determined by two-tailed unpaired Student's *t* test. **P* < 0.05; ns, not significant.

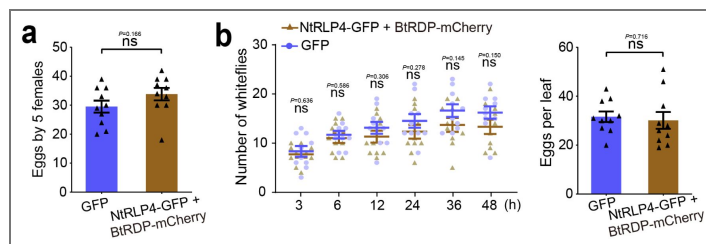


Fig. S20. Degradation of NtRNP4 by purified BtRDP^{SP} in *Nicotiana benthamiana* leaves.

Recombinant BtRDP^{SP}-his and GFP-his proteins are expressed in *Escherichia coli*, and purified by Ni-NTA. *N. benthamiana* plants overexpressing NtRNP4-myc or NtCf9-myc are then infiltrated with different concentrations of purified BtRDP^{SP}-his and GFP-his. The samples are probed with anti-myc antibodies for immunoblot analysis. Rubisco staining (RbcL) is conducted to visualize the amount of sample loading. Coomassie brilliant blue (CBB) staining is conducted to visualize the amount of recombinant BtRDP^{SP}-his and GFP-his proteins. Experiments are repeated three times with the similar results. Band density is measured using ImageJ. The density values from three biological replicates are calculated and the mean value in the first lane is set at 1.0. Data are presented as mean values $\pm$ SEM.

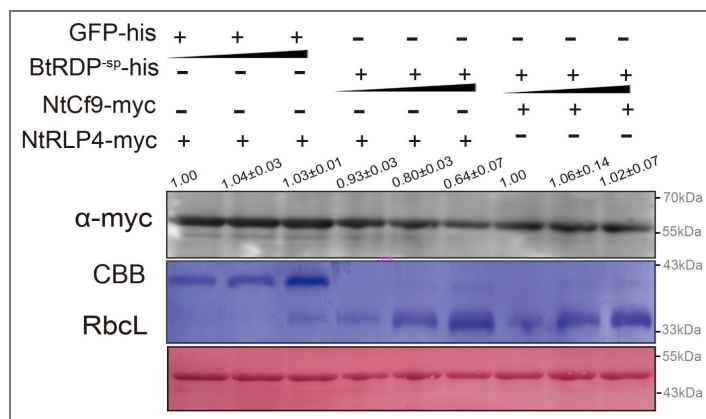


Fig. S21. Effect of BtRDP on NtRPL4 and NtSOBIR1.

NtRPL4-myc and NtSOBIR1-flag were transiently co-expressed with different concentration of BtRDP-flag in *Nicotiana benthamiana* plants through *Agrobacterium* infiltration. The samples are probed with anti-flag and anti-myc antibodies for immunoblot analysis. Rubisco staining (RbCL) is conducted to visualize the amount of sample loading. The amount of BtRDP were controlled by infiltrating different concentrations of *Agrobacterium* ($OD_{600} = 0.1, 0.3, \text{ and } 1.0$).

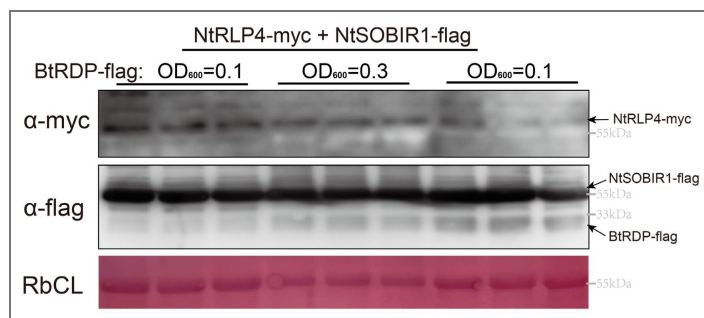


Fig. S22. Effect of BtRDP on the transcript level of NtRPL4.

NtRPL4-myc is agroinfiltrated together with different concentration of BtRDP-flag or GFP-flag into *Nicotiana benthamiana* leaves. The transcript level of *NtRPL4* is determined by qRT-PCR. Data are presented as mean values $\pm$ SEM. Three independent biological replicates are performed. The same lowercase letters "a" indicate no statistically significant differences at $P < 0.05$ according to the one-way ANOVA test followed by Tukey's multiple comparisons test.

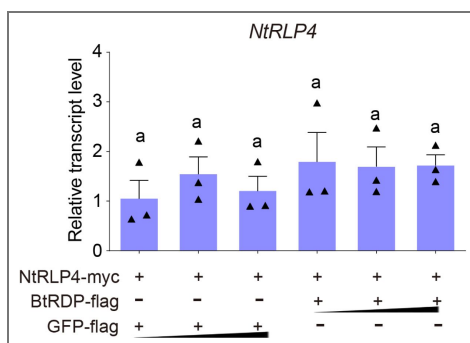


Fig. S23. Effects of autophagy inhibitor on NtRFP4 accumulation.

NtRFP4-myc and BtRDP-flag were transiently co-expressed in *Nicotiana benthamiana* plants through *Agrobacterium* infiltration. Co-infiltrated leaves are treated with autophagy inhibitor BAF and 3-MA at 24 h post injection. The samples are probed with anti-flag and anti-myc antibodies for immunoblot analysis. Rubisco staining (RbCL) is conducted to visualize the amount of sample loading. The small triangle indicates the different concentrations of *Agrobacterium* ($OD_{600} = 0.1$ and 1.0). Experiments are repeated three times with the similar results. Band density is measured using ImageJ. The density values from three biological replicates are calculated and the mean value in the first lane is set at 1.0 . Data are presented as mean values $\pm$ SEM. *P*-values are determined by two-tailed unpaired Student's *t* test. ****P* < 0.001 ; ***P* < 0.01 ; **P* < 0.05 ; ns, not significant.

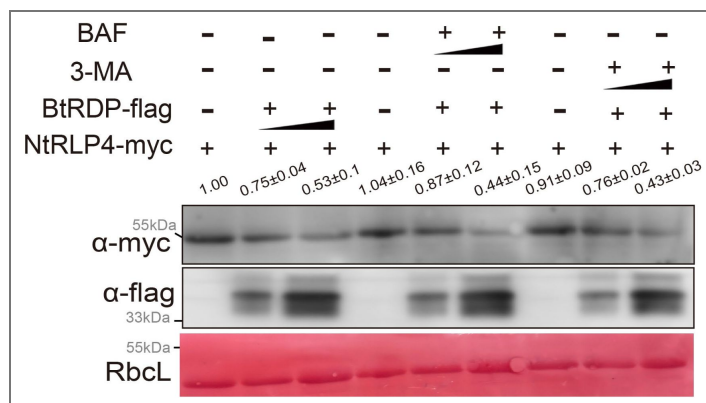


Fig. S24. Sequence alignment and phylogenetic tree of RLP4 homologs.

(a) The amino acid sequence of RLP4s from *Nicotiana tabacum*, *N. benthamiana*, *Solanum lycopersicum*, and *Oryza sativa* are aligned using ClusterX software. Black shades indicate the conserved regions. (b) Phylogenetic analysis of RLP4 and Cf9-associated genes used in this study. The unrooted phylogenetic trees are constructed by RAxML v0.9.0 using the maximum likelihood method with 1000 bootstrap replicates.

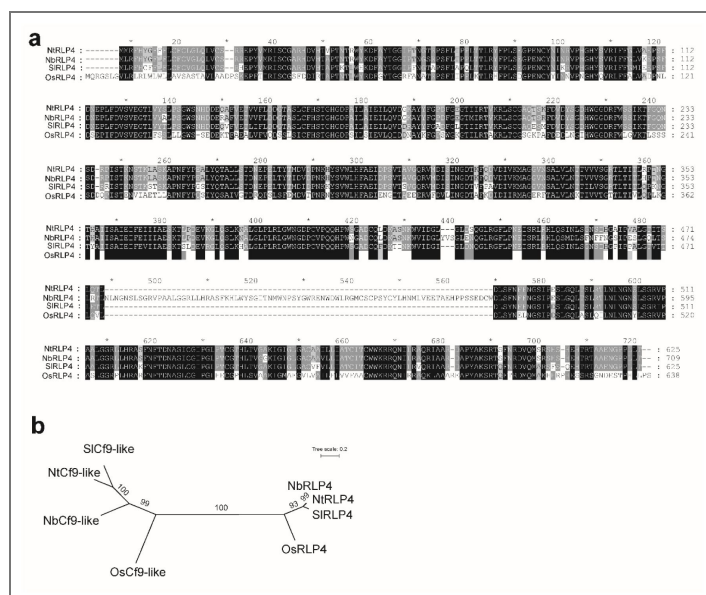
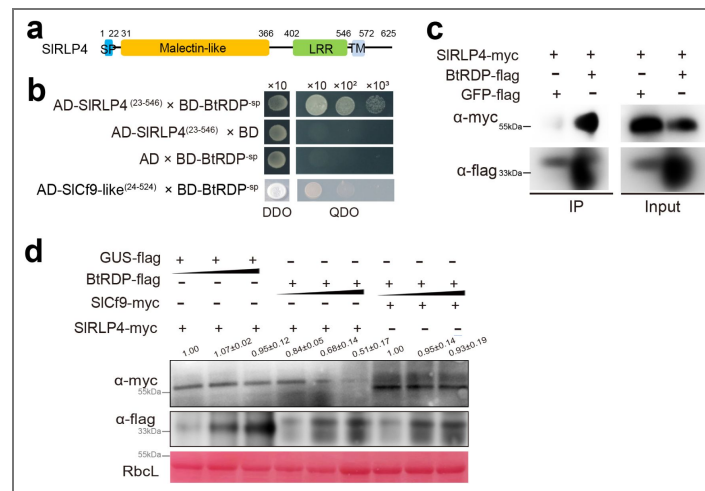


Fig. S25. Influence of BtRDP on RLP4 homology in *Solanum lycopersicum*.

(a) Domain organization of *S. lycopersicum* RLP4 (SIRLP4). SIRLP4 contains a predicted N-terminal signal peptide (SP), a malectin-like domain, a LRR domain, and a transmembrane (TM) domain. (b, c) Yeast two-hybrid and co-immunoprecipitation (Co-IP) assays showing the interaction between BtRDP and SIRLP4. In (b), BtRDP is expressed without a signal peptide (BtRDP^{-SP}), while SIRLP4 and SICf9 are expressed without a signal peptide and transmembrane domain (SIRLP4⁽²³⁻⁵⁴⁶⁾ and SICf9⁽²⁴⁻⁵²⁴⁾). (d) Effect of BtRDP on the accumulation of SIRLP4. SIRLP4-myc and SICf9-myc are agro-injected together with flag or myc tags at C-terminal ends, respectively. The small triangle indicates the different concentrations (OD₆₀₀ = 0.05, 0.3, and 1.0) of *Agrobacterium*. Rubisco staining (Rbcl) is conducted to visualize the amount of sample loading. Experiments are repeated three times with the similar results. Band density is measured using ImageJ. The density values from three biological replicates are calculated. For SIRLP4-myc bands (lane 1-6), the lane 1 is set at 1.0. For SICf9-myc bands (lane 7-9), the lane 7 is set at 1.0.

**Fig. S26. Yeast two hybrid assays showing the interaction between OsRLP4 and salivary proteins from planthopper species.**

OsRLP4₍₂₉₋₅₅₁₎ (OsRLP4 without signal peptides and transmembrane domains) is fused to pGADT7 vector, while salivary proteins without signal peptides are fused to pGBKT7 vector, respectively. The different combinations of constructs are transformed into yeast cells, and are grown on the selective medium SD/-Trp/-Leu (DDO), and the interactions are tested with SD/-Trp/-Leu/-His/-Ade (QDO). Twenty salivary proteins are selected. The accession number of each protein is listed as follow: (1) MF278694.1, (2) XP_039291719.1, (3) KU365967.1, (4) MF278706.1, (5) MF278711.1, (6) MF278714.1, (7) MF278715.1, (8) MF278720.1, (9) XP_022195702.1, (10) XP_022196818.2, (11) XP_039284081.1, (12) XP_022192221.2, (13) XP_022207944.2, (14) KT764973.1, (15) RZF44823.1, (16) RZF42644.1, (17) RZF33006.1, (18) RZF48570.1, (19) RZF42817.1, (20) RZF33751.1.

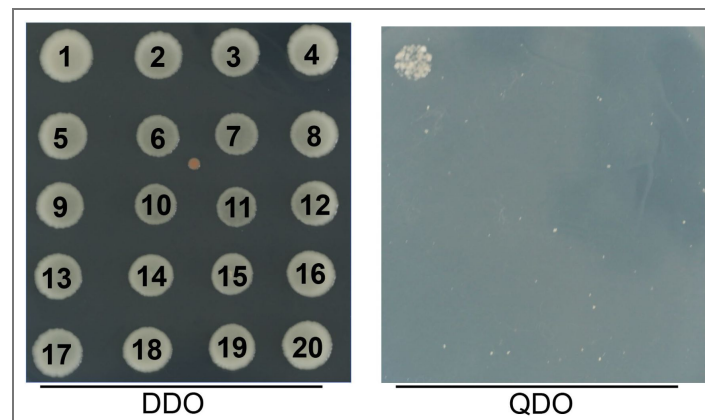


Fig. S27. Three-dimensional structure of BtRDP and NISP104.

AlphaFold2 is used to predicate the protein structure. The structures are colored as a spectrum from N-terminus (blue) to C-terminus (red).

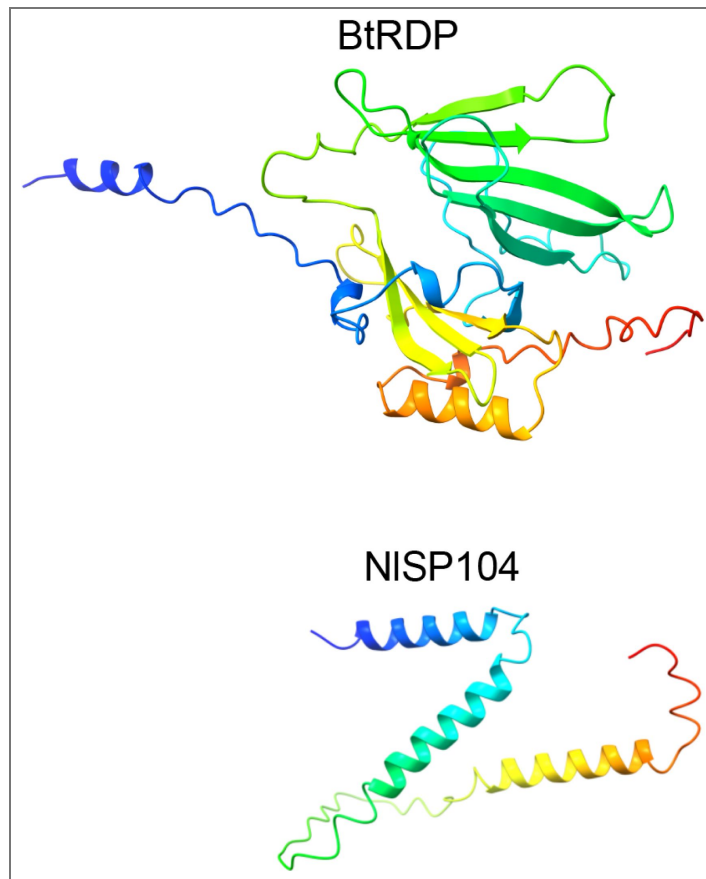
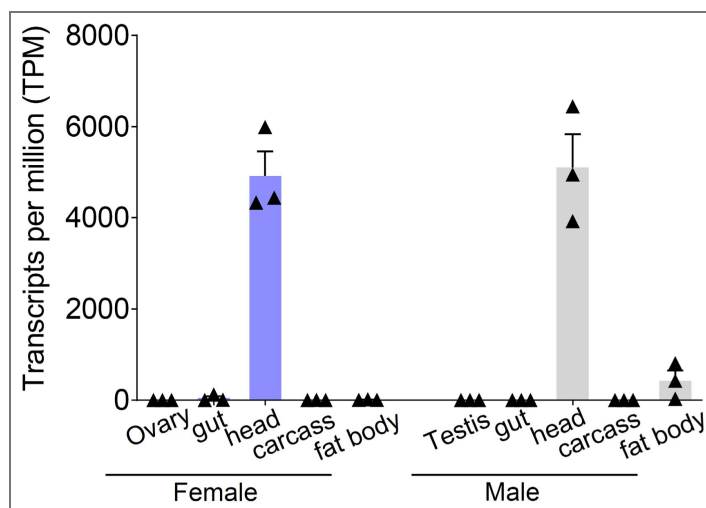


Fig. S28. Expression patterns of *Nilaparvata lugens* NISP104.

Transcripts per million (TPM) expression values of *NISP104* in different tissues are determined based on the transcriptomic data. The head that contains salivary glands exhibits highest *NISP104* expression in male and female *N. lugens*. Data are presented as mean values $\pm$ SEM (n=3 independent biological replicates).



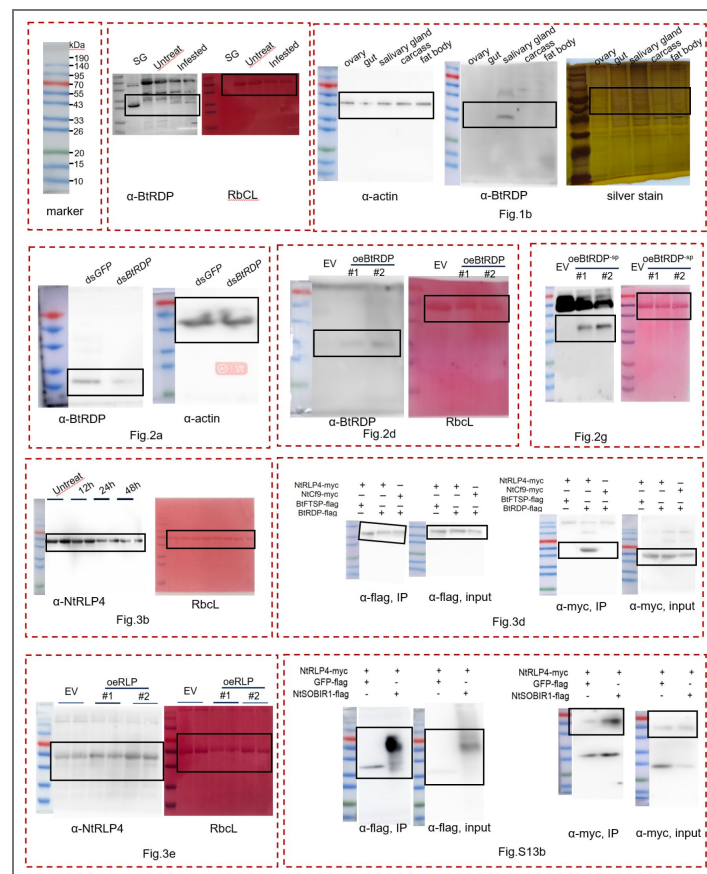


Fig. S29. Original images for blots in Figures 1-3 and Supplementary Figure S13b.

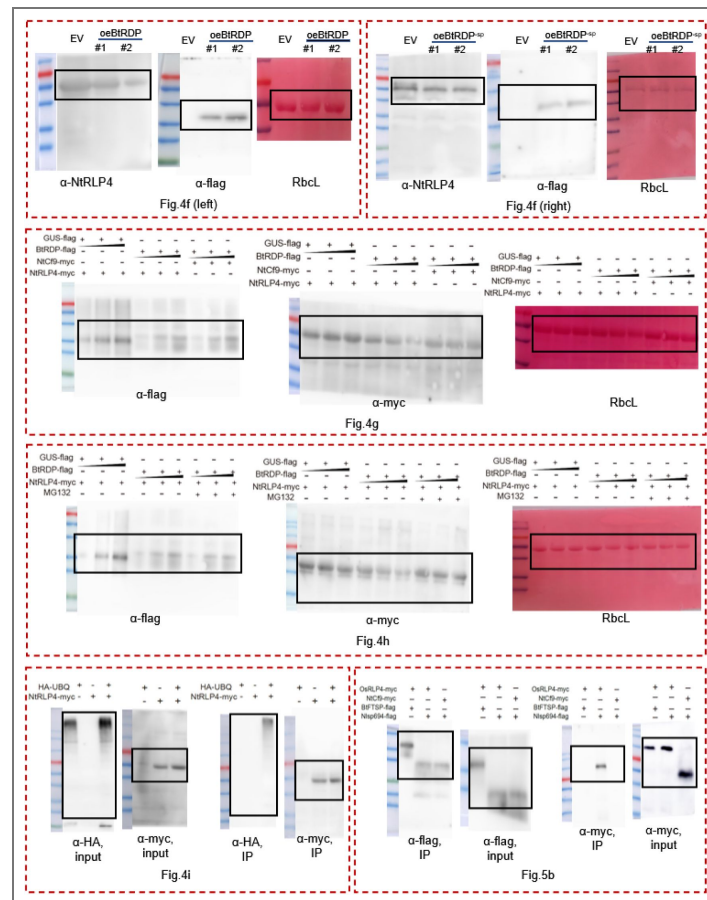


Fig. S30. Original images for blots in [Figures 4b-5b](#).

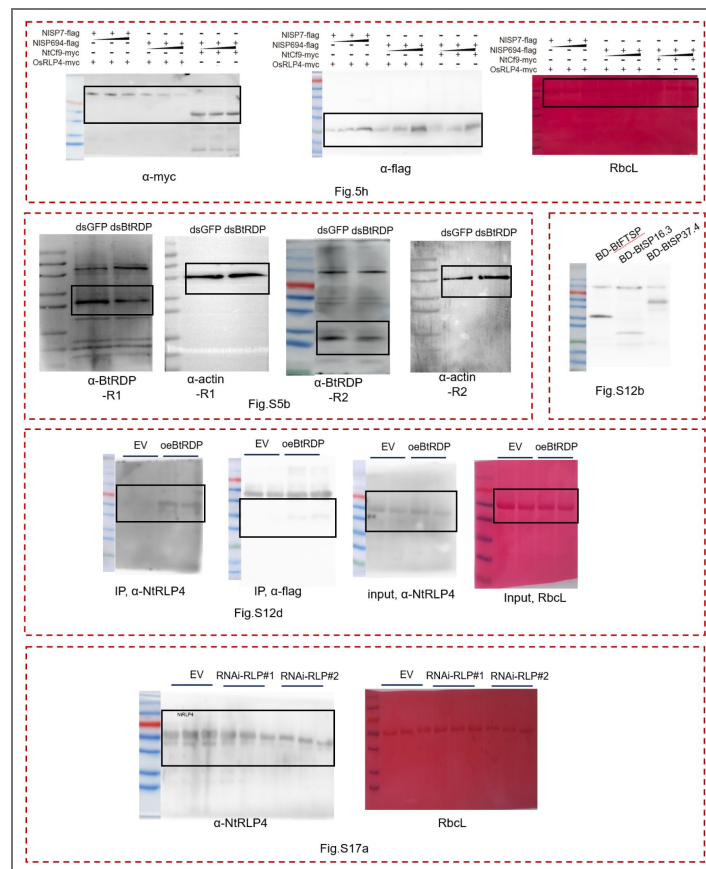


Fig. S31. Original images for blots in [Figures 5](#) and [Supplementary Figure 5-17](#).

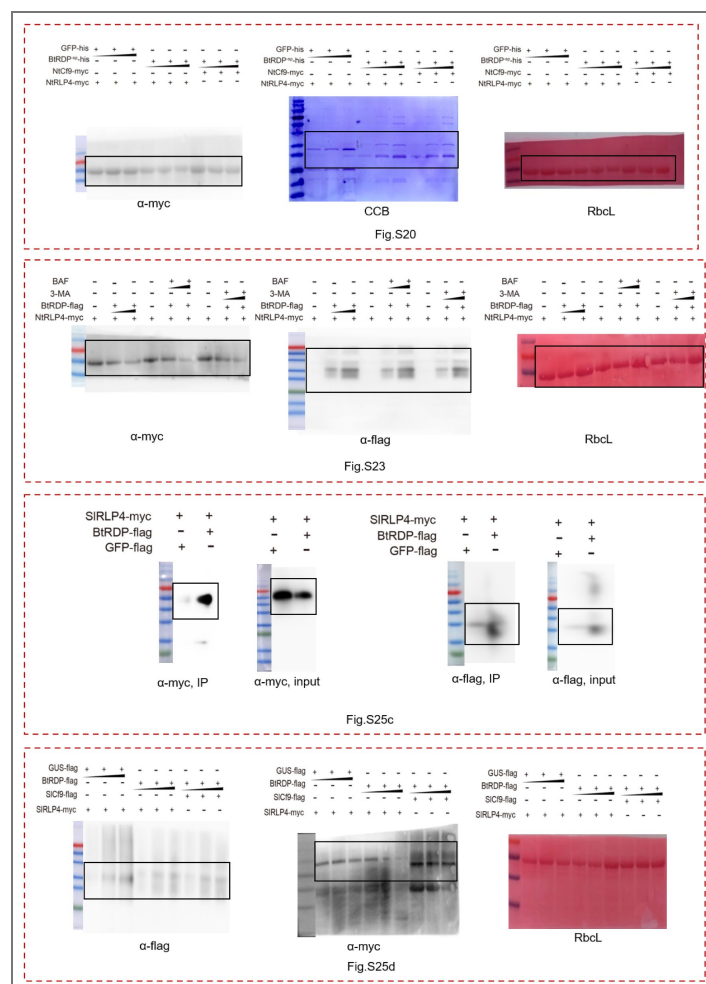


Fig. S32. Original images for blots in Supplementary Figure 20-25.

Supplementary Tables

Family	Species	Accession	Total Base	Number of RDP homologs	Number of SP101 homologs	Note
Aleyrodidae	<i>Bemisia tabaci</i> MED	GCA_918797505	609.6 MB	1	0	Assembled genome
	<i>Bemisia tabaci</i> MEAM1	GCF_001854935	600.6 MB	1	0	Assembled genome
	<i>Bemisia tabaci</i> India	SRR1159209	9.7 GB*	1	0	Assembled genome
	<i>Bemisia tabaci</i> SSA1	GCA_902825415	657.8 MB	1	0	Assembled genome
	<i>Bemisia tabaci</i> SSA2	GCA_903994125	625.3 MB	1	0	Assembled genome
	<i>Aleurocanthus spiniferus</i>	SRR17330024	9.2 GB*	1	0	RNA-seq, whole body
	<i>Aleyrodes proletella</i>	SRX14998536	1.9 GB*	1	0	RNA-Seq, whole body
	<i>Trialeurodes vaporariorum</i>	GCA_011764245	814.7 MB	1	0	Assembled genome
	<i>Dialeurodes citri</i>	SRR2980521	5.0 GB*	1	0	RNA-Seq, whole body
Delphacidae	<i>Nilaparvata lugens</i>	GCF_014356525	1,088 MB	0	1	Assembled genome
	<i>Laodelphax striatellus</i>	GCA_017141395	510.2 MB	0	1	Assembled genome
	<i>Sogatella furcifera</i>	GCA_017141385	563.8 MB	0	1	Assembled genome
Aphididae	<i>Acyrtosiphon pisum</i>	GCA_005508785	533.6 MB	0	0	Assembled genome
	<i>Myzus persicae</i>	GCA_001856785	347.3 MB	0	0	Assembled genome
	<i>Sitobion miscanthi</i>	GCA_008086715	397.9 MB	0	0	Assembled genome
Aphalaridae	<i>Pachypsylla venusta</i>	GCA_012654025	482.0 MB	0	0	Assembled genome
	<i>Diaci psyllid</i>	GCA_000475195	485.7 MB	0	0	Assembled genome
Alydidae	<i>Riptortus pedestris</i>	GCA_019009955	1.0 GB	0	0	Assembled genome
Lygaeidae	<i>Oncopeltus fasciatus</i>	GCA_000696205	1.1 GB	0	0	Assembled genome
Pentatomidae	<i>Halyomorpha halys</i>	GCA_000696795	998.2 MB	0	0	Assembled genome
Cimicidae	<i>Cimex lectularius</i>	GCA_000648675	510.8 MB	0	0	Assembled genome
Reduviidae	<i>Rhodnius prolixus</i>	GCA_000181055	706.8 MB	0	0	Assembled genome
Gerridae	<i>Gerris buenoi</i>	GCA_001010745	994.4 MB	0	0	Assembled genome
Drosophilidae	<i>Drosophila melanogaster</i>	GCA_002310755	120.4 MB	0	0	Assembled genome
Culicidae	<i>Aedes albopictus</i>	GCA_018104305	1.3 GB	0	0	Assembled genome
Tenebrionidae	<i>Tribolium castaneum</i>	GCA_000002335	165.9 MB	0	0	Assembled genome
Chrysomeloidea	<i>Anoplophora glabripennis</i>	GCA_000390285	679.8 MB	0	0	Assembled genome
Bombycidae	<i>Bombyx mori</i>	GCA_027366755	456.6 MB	0	0	Assembled genome
Noctuidae	<i>Spodoptera frugiperda</i>	GCA_023101765	383.9 MB	0	0	Assembled genome
Pediculidae	<i>Pediculus humanus</i>	GCA_000006295	110.8 MB	0	0	Assembled genome

* The total base of raw data retrieved from SRA database.

Table S1. Identification of RDP and SP101 homologs in insect species

<i>N. benthamiana</i> v2.6.1 Accession ¹	Annotation	Number of colonies	Region
Niben261Chr07g1310001.1	Putative leucine-rich repeat receptor-like serine/threonine-protein kinase	3	455-709
Niben261Chr13g1396009.1	Chloroplast stem-loop binding protein of 41 kDa b	1	162-374
Niben261Chr02g0086007.1	Sorbitol dehydrogenase-like	1	95-305
Niben261Chr17g0179002.1	Guanine nucleotide-binding protein subunit beta-like protein	1	92-326
Niben261Chr01g1673009.1	Ribulose biphosphate carboxylase/oxygenase activase 2	2	228-420
Niben261Chr07g1128067.1	Ribulose-1,5-bisphosphate carboxylase/oxygenase small subunit	5	23-181
Niben261Chr14g0161005.1	Cytochrome b6-f complex iron-sulfur subunit 2	5	1-216

¹ The sequence can be downloaded from Sol Genomics Network (https://solgenomics.net/ftp/genomes/Nicotiana_benthamianaV261)

Table S2. Proteins from a *Nicotiana benthamiana* cDNA library screened by yeast two hybrid using BtRDP as a bait

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Additional information

Data availability

All data is included in the manuscript and/or supporting information. The transcriptomic data have been submitted to the National Genomics Data Center under accession number PRJCA025907. The original blot images have been displayed in Fig. S29 [S29](#)-S32 [S32](#).

Author Contributions Statement

H-JH, J-ML, C-XZ, and J-PC planned and designed the research. XW, J-BL, Y-ZW, and H-JH performed experiments and analyzed data. X-HZ provided the EV plants. H-JH and XW drafted the manuscript.

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MOST National Natural Science Foundation of China (NSFC)	U23A6006	Hai-Jian Huang
MOST NSFC NSFC-Zhejiang Joint Fund 浙江省科学技术厅 Natural Science Foundation of Zhejiang Province (ZJNSF)	LDQ24C140001	Hai-Jian Huang

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Additional files

Table S3. [Differentially expressed genes between empty vector \(EV\) and oeRLP#1 transgenic plant.](#)

Table S4. [Primers used in this study.](#)

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Peer reviews

Reviewer #1 (Public review):

Summary:

This manuscript investigates how herbivorous insects, specifically whiteflies and planthoppers, utilize salivary effectors to overcome plant immunity by targeting the RLP4 receptor.

Strengths:

The authors present a strong case for the independent evolution of these effectors and provide compelling evidence for their functional roles.

<https://doi.org/10.7554/eLife.108737.2.sa3>

Reviewer #2 (Public review):

Summary:

The authors tested an interesting hypothesis that white flies and planthoppers independently evolved salivary proteins to dampen plant immunity by targeting a receptor-like protein. Unlike previously reported receptor-like proteins with large ligand-binding domains, the NtRLP4 here has a malectin LRR domain. Interestingly, it also associates with the adaptor SOBIR1. While the function of this protein remains to be further explored, the authors provide strong evidence showing it's the target of salivary proteins as the insects' survival strategy.

Major points:

The authors mixed the concepts of LRR-RLPs with malectin LRR-RLPs. These are two different type of receptors. While LRR-RLPs are well studied, little is known about malectin LRR-RLPs. The authors should not simply apply the mode of function of LRR-RLPs to RLP4 which is a malectin LRR-RLP. In addition, LRR-RLPs that function as ligand-binding receptors typically possess >20 LRRs, whereas RLP4 in this work has a rather small ectodomain. It remains unclear whether it will function as a PRR.

I can't agree with the author's logic of testing uninfested plants for proving a PRR's function. The function of a pattern recognition receptor depends on perceiving the corresponding ligand. As shown by the data provided, RLP4-OE plants have altered transcriptional profile

indicating activated defense, suggesting it's unlikely a PRR. An alternative explanation is needed.

More work on BAK1 will also help to clarify the ideas proposed by the authors.

<https://doi.org/10.7554/eLife.108737.2.sa2>

Reviewer #3 (Public review):

Summary:

In this study, Wang et al., investigate how herbivorous insects overcome plant receptor-mediated immunity by targeting plant receptor-like proteins. The authors identify two independently evolved salivary effectors, BtRDP in whiteflies and NlSP694 in brown planthoppers, that promote the degradation of plant RLP4 through the ubiquitin-dependent proteasome pathway. NtRLP4 from tobacco and OsRLP4 from rice are shown to confer resistance against herbivores by activating defense signaling, while BtRDP and NlSP694 suppress these defenses by destabilizing RLP4 proteins.

Strengths:

This work highlights a convergent evolutionary strategy in distinct insect lineages and advances our understanding of insect-plant coevolution at the molecular level.

Two minor comments:

In line 140, yeast two-hybrid (Y2H) was used to screen for interacting proteins in plants. However, it is generally difficult to identify membrane receptors using Y2H. Please provide more methodological details to justify this approach, or alternatively, include a discussion explaining this.

In Figure S12C, the interaction between the two proteins appears to be present in the nucleus as well. Please provide a possible explanation for this observation.

<https://doi.org/10.7554/eLife.108737.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary:

This is a well-structured and interesting manuscript that investigates how herbivorous insects, specifically whiteflies and planthoppers, utilize salivary effectors to overcome plant immunity by targeting the RLP4 receptor.

Strengths:

The authors present a strong case for the independent evolution of these effectors and provide compelling evidence for their functional roles.

Weaknesses:

Western blot evidence for effector secretion is weak. The possibility of contamination from insect tissues during the sample preparation should be avoided.

Below are some specific comments and suggestions to strengthen the manuscript.

Thank you very much for your comments. We have carefully revised the MS following your valuable suggestions and comments.

(1) Western blot evidence for effector secretion:

The western blot evidence in Figure 1, which aims to show that the insect protein is secreted into plants, is not fully convincing. The band of the expected size (~30 kDa) in the infested tissues is very weak. Furthermore, the high and low molecular weight bands that appear in the infested tissues do not match the size of the protein in the insects themselves, and a high molecular weight band also appears in the uninfested control tissues. It is difficult to draw a definitive conclusion that this protein is secreted into the plants based on this evidence. The authors should also address the possibility of contamination from insect tissues during the sample preparation and explain how they have excluded this possibility.

Thank you for pointing out this. One or two bands between 25-35kDa were specifically identified in *B. tabaci*-infested plants, but not the non-infested plants, and the smaller high intensity band is the same size as that of BtRDP in salivary glands. This experiment has been repeated for six times. In the current version, we reperformed this experiment, and provided salivary gland sample as a positive control, which showed the same molecular weight with a specific band in infested sample. It is noteworthy that in the experiment of current version, only the smaller high intensity band appear, while the low intensity band did not appear. The detection of a protein within infested plant tissue is a key criterion for validating the secretion of salivary effectors, an approach supported by numerous studies in this field. Furthermore, our previous LC-MS/MS analysis of *B. tabaci* watery saliva identified six unique peptides matching BtRDP, providing independent evidence for its presence in saliva. Therefore, as we now state in the manuscript “the detection of BtRDP in infested plants (Fig. 1a) and in watery saliva (Fig. S1) collectively indicates that BtRDP is a salivary protein”.

Regarding the higher molecular weight band that present in both infested and non-infested samples, we agree that it most likely represents a non-specific band, which is a common occurrence in Western blot assays. Such bands are sometimes used to indicate comparable sample loading. To address the possibility of contamination by insect tissues, we wish to clarify that all insects and deposited eggs were carefully removed from the infested leaves prior to sample processing. Moreover, BtRDP is undetectable at the egg stage, and no BtRDP-associated band can be detected even in egg contamination. We have revised the Methods section to explicitly state this procedure:

“After feeding, the eggs deposited on the infested tobacco leaves were removed. The leaves showing no visible insect contamination were immediately frozen in liquid nitrogen and ground to a fine powder.”

(2) Inconsistent conclusion (Line 156 and Figure 3c):

The statement in line 156 is inconsistent with the data presented in Figure 3c. The figure clearly shows that the LRR domain of the protein is the one responsible for the interaction with BtRDP, not the region mentioned in the text. This is a critical misrepresentation of the experimental findings and must be corrected. The conclusion in the text should accurately reflect the data from the figure.

We apologize for any confusion caused by the original phrasing. In our previous manuscript, the description “NtRLP4 without signal peptides and transmembrane domains” referred specifically to the truncated construct NtRLP4₍₂₃₋₅₄₁₎ used in the experiment. To prevent any misunderstanding, we have revised the sentence in the updated version to state explicitly:

“Point-to-point Y2H assays reveal that NtRPL4₍₂₃₋₅₄₁₎ (a truncated version lacking the signal peptide and transmembrane domains) interacts with BtRDP^{SP}”.

(3) Role of SOBIR1 in the RLP4/SOBIR1 Complex:

The authors demonstrate that the salivary effectors destabilize the RLP4 receptor, leading to a decrease in its protein levels and a reduction in the RLP4/SOBIR1 complex. A key question remains regarding the fate of SOBIR1 within this complex. The authors should clarify what happens to the SOBIR1 protein after the destabilization of RLP4. Does SOBIR1 become unbound, targeted for degradation itself, or does it simply lose its function without RLP4? This would provide further insight into the mechanism of action of the effectors.

Thank you for suggestion. In the current version, we assessed the impact of BtRDP on NtSOBIR1 following NtRPL4 destabilization. The results showed that while the NtRPL4-myc accumulation was markedly reduced, NtSOBIR1-flag levels remained unchanged, suggesting that destabilization of NtRPL4 did not affect NtSOBIR1 accumulation.

(4) Clarification on specificity and evolutionary claims:

The paper's most significant claim is that the effectors from both whiteflies and planthoppers "independently evolved" to target RLP4. While the functional data is compelling, this evolutionary claim would be more convincing with stronger evidence. Showing that two different effector proteins target the same host protein is a fascinating finding but without a robust phylogenetic analysis, the claim of independent evolution is not fully supported. It would be valuable to provide a more detailed evolutionary analysis, such as a phylogenetic tree of the effector proteins, showing their relationship to other known insect proteins, to definitively rule out a shared, but highly divergent, common ancestor.

We appreciate the reviewer's valuable suggestion to investigate a potential evolutionary link between BtRDP and NtSP104. Our initial analysis already indicated no detectable sequence similarity. To address this point more thoroughly, we attempted a phylogenetic analysis. However, we were unable to generate a meaningful alignment due to a complete lack of conserved amino acid sequences. Therefore, we conducted a comparative genomics analysis by blasting both proteins against the genomic or transcriptomic data of 30 diverse insect species. This analysis revealed that RDP is exclusively present in Aleyrodidae species, and SP104 is exclusively present in Delphacidae species (Table S1). Taken together, the absence of sequence similarity, their distinct protein structure, and their lineage-specific distributions, we conclude that BtRDP and NtSP104 are highly unlikely to be homologous and thus did not originate from a common ancestor.

(5) Role of SOBIR1 in the interaction:

The results suggest that the effectors disrupt the RLP4/SOBIR1 complex. It is not entirely clear if the effectors are specifically targeting RLP4, SOBIR1, or both. Further experiments, such as a co-immunoprecipitation assay with just RLP4 and the effector, could clarify if the effector can bind to RLP4 in the absence of SOBIR1. This would help to definitively place RLP4 as the primary target.

We appreciate the reviewer's insightful comments regarding whether the effector preferentially targets RLP4, SOBIR1, or both. In our study, we conducted reciprocal co-immunoprecipitation assays using RLP4 and BtRDP as controls. These assays showed that BtRDP interacts with RLP4 but does not interact with SOBIR1, supporting the conclusion that SOBIR1 is unlikely to be a direct target of BtRDP. We fully agree that testing the interaction between RLP4 and BtRDP in the absence of SOBIR1 would further strengthen the conclusion. However, we were unable to obtain *N. tabacum* SOBIR1 knockout mutants, and therefore

could not experimentally assess whether the RLP4–BtRDP interaction persists in planta without SOBIR1. Nevertheless, our yeast two-hybrid assays demonstrate that RLP4 and BtRDP can directly interact, indicating that their association does not strictly depend on SOBIR1. Together, these results support the interpretation that RLP4 is the primary target of BtRDP, while SOBIR1 is not directly engaged by the effector.

(6) Transcriptome analysis (Lines 130-143):

The transcriptome analysis section feels disconnected from the rest of the manuscript. The findings, or lack thereof, from this analysis do not seem to be directly linked to the other major conclusions of the paper. This section could be removed to improve the manuscript's overall focus and flow. If the authors believe this data is critical, they should more clearly and explicitly connect the conclusions of the transcriptome analysis to the core findings about the effector-RLP4 interaction.

Thank you for suggestion. As you and Reviewer #2 pointed, the transcriptomic analysis did not closely link to the major conclusions of the paper, and we got little information from the transcriptomic analysis. Therefore, we remove these analyses to improve the manuscript's overall focus and flow.

(7) Signal peptide experiments (Lines 145 and beyond):

The experiments conducted with the signal peptide (SP) are questionable. The SP is typically cleaved before the protein reaches its final destination. As such, conducting experiments with the SP attached to the protein may have produced biased observations and could lead to unjustified conclusions about the protein's function within the plant cell. We suggest the authors remove the experiments that include the signal peptide.

Thank you for pointing out this. The SP was retained to direct the target proteins to the extracellular space of plant cells. Theoretically, the SP is cleaved in the mature protein. This methodology is widely used in effector biology. For example, the SP directs *Meloidogyne graminicola* Mg01965 to the apoplast, where it functions in immune suppression, whereas Mg01965 without the SP fails to exert this function (10.1111/mpp.12759). In our study, the SP of BtRDP was expected to guide the target protein to the extracellular space, facilitating its interaction with RLP4. Moreover, the observed protein sizes of BtRDP with and without the SP in transgenic plants were identical, suggesting successful SP cleavage. Therefore, we have retained the experiments involving the SP in the current version.

(8) Overly strong conclusion and unclear evidence (Line 176):

The use of the word "must" on line 176 is very strong and presents a definitive conclusion without sufficient evidence. The authors state that the proteins must interact with SOBIR1, but they do not provide a clear justification for this claim. Is SOBIR1 the only interaction partner for NtRLP4? The authors should provide a specific reason for focusing on SOBIR1 instead of demonstrating an interaction with NtRLP4 first. Additionally, do BtRDP or NISP694 also interact with SOBIR1 directly? The authors should either tone down their language to reflect the evidence or provide a clearer justification for this strong claim.

Thank you for pointing this out. In the current version, the word “must” has been toned down to “may” due to insufficient supporting evidence. In this study, SOBIR1 was chosen because it has been widely reported to be required for the function of several RLPs involved in innate immunity. However, it remains unclear whether SOBIR1 is the only interaction partner of NtRLP4. In the current version, we have clarified the rationale for focusing on SOBIR1 prior to the experiments “The receptor-like kinase SOBIR1, which contains a kinase domain, has been widely reported to be required for the function of RLPs involved in innate immunity (Gust & Felix, 2014)” and discussed that “Although NtRLP4 interacts with SOBIR1, this alone

does not confirm that it operates strictly through this canonical module. Evidence from other RLPs shows that co-receptor usage can be flexible, and some RLPs function partly or conditionally independent of SOBIR1. Therefore, a more definitive assessment of NtRLP4 signaling will therefore require genetic dissection of its co-receptor dependencies, including but not limited to SOBIR1.” In addition, the direct interaction between BtRDP and SOBIR1 was experimentally tested, and the results showed that BtRDP failed to interact with SOBIR1.

Minor Comments

(9) The statement in the abstract, "However, it remains unclear how these invaders are able to overcome receptor perception and disable the plant signaling pathways," is not entirely accurate. The fields of effector biology and host-pathogen interactions have provided significant insight into how pathogens and pests manipulate both Pattern-Triggered Immunity (PTI) and Effector-Triggered Immunity (ETI). While the specific mechanism described in this paper is novel, the broader claim that the field is unclear on these processes weakens the initial hook of the paper. A more precise framing of the problem would be beneficial, perhaps by stating that the specific mechanisms used by these particular herbivores to target RLP4 were previously unknown.

Thank you for this insightful comment. We agree that the original statement in the abstract overstated the lack of understanding in the field. In the current version, we have refined the sentence to more accurately reflect the current state of knowledge, emphasizing that while microbial suppression of plant immunity has been extensively studied, the strategies used by herbivorous insects to overcome receptor-mediated defenses remain less understood. The revised sentence now reads as follows: "Although the mechanisms used by microbial pathogens to suppress plant immunity are well studied, how herbivorous insects overcome receptor-mediated defenses remains unclear".

(10) The introduction is heavily focused on Pattern Recognition Receptors (PRRs), which, while central to the paper's findings, gives a somewhat narrow view of the plant's defense against herbivores. It would be beneficial to briefly acknowledge the broader context of plant defenses, such as physical barriers, direct chemical toxicity, and indirect defenses, before narrowing the focus to the specific molecular interactions of PRRs that are the core of this study. This would provide a more complete picture of the "arms race" between plants and herbivores.

Thank you for this valuable suggestion. We agree that the original introduction focused too narrowly on pattern-recognition receptors (PRRs). In the current version, we have expanded the introductory section to provide a broader overview of plant defense mechanisms. Specifically, we now acknowledge the multiple layers of plant defenses, including physical barriers (e.g., cuticle and cell wall), chemical defenses (e.g., toxic secondary metabolites and anti-nutritive compounds), and indirect defenses mediated by herbivore-induced volatiles. This addition provides a more complete context for understanding the molecular interactions discussed in this study. The revised paragraph now reads as follows: "Plants have evolved sophisticated defense systems to survive constant attacks from pathogens and herbivorous insects. These defenses operate at multiple levels, including physical barriers such as the cuticle and cell wall, chemical defenses involving toxic secondary metabolites and anti-nutritive compounds, and indirect defenses that attract natural enemies of herbivores through the emission of herbivore-induced volatiles. Beyond these general strategies, plants also rely on highly specialized molecular immune responses that allow them to detect and respond rapidly to invaders."

(11) The figure legends are generally clear, but some could be more detailed. For instance, in Figure 2, it would be helpful to explicitly state what each bar represents in the graph and to include the statistical test used. Please ensure all panels in all figures have clear labels.

Thank you for this helpful suggestion. We have revised the legend of Fig. 2 and other figures to provide more detailed information for each panel. Specifically, we now explicitly describe what each bar represents in the graphs and specify the statistical test used. In addition, we ensured that all panels are clearly labeled. These changes improve clarity and allow readers to better interpret the data.

(12) The methods section is comprehensive, but it would be helpful to include more specifics on the statistical analyses used. For example, the type of statistical test (e.g., t-test, ANOVA) and the software used should be mentioned for each experiment.

Thank you for your suggestion. We have revised the Methods section (Statistical analysis) to provide more detailed information on the statistical analysis used for each experiment.

(13) The manuscript's overall impact is weakened by the inclusion of unnecessary words and a few grammatical issues. A focused revision to tighten the language would make the major findings stand out more clearly. For example, on page 2, line 18, "in whitefly Bemisia tabaci, BtRDP is an Aleyrod..." seems to have an incomplete sentence. A thorough proofreading for typos and grammatical errors is highly recommended to improve the overall readability.

Thank you for your suggestion. We have carefully revised the abstract and the manuscript to improve clarity, readability, and grammatical correctness. In addition, we sought the assistance of a professional English editor to thoroughly proofread and polish the manuscript, ensuring that the language meets high academic standards.

(14) The discussion section is strong, but it could benefit from a more explicit connection between the findings and the broader ecological implications. For instance, how might the independent evolution of these effectors in different insect species impact plant-insect co-evolutionary dynamics?

We thank the reviewer for the valuable suggestion. In the current version, we have added a paragraph in the Discussion section highlighting the broader ecological and evolutionary implications of our findings. Specifically, we discuss how the independent evolution of RLP4-targeting effectors in different insect lineages may drive plant-insect co-evolution, influence selection pressures on both plants and herbivores, and potentially shape defense diversification across plant communities. This addition helps to link our molecular findings to ecological outcomes and co-evolutionary dynamics.

(15) The sentence on line 98, which reads "A few salivary proteins have been reported to attach to salivary sheath after secretion" seems to serve an unclear purpose in the introduction. It would be helpful for the authors to clarify its relevance to the surrounding context or to the paper's overall argument. Its inclusion currently disrupts the flow of the introduction and makes it difficult for the reader to understand its intended purpose.

We thank the reviewer for the comment. We have revised the paragraph to clarify the relevance of salivary sheath localization to the study. Specifically, we now introduce the role of the salivary sheath as a potential scaffold for effector delivery and explicitly link previous reports of sheath-associated salivary proteins to our observation that BtRDP localizes to the salivary sheath after secretion.

(16) The writing in lines 104-106 is both grammatically inconsistent and overly wordy. The authors switch between present and past tense ("is" and "was"), and the sentences could be made more concise to improve the clarity and flow of the text. Also check entire paper.

We thank the reviewer for pointing this out. We have revised the sentence to improve grammatical consistency and clarity, and also checked the manuscript for similar issues. The sentence is now split into two concise statements. In addition, we have thoroughly checked the entire manuscript for similar tense inconsistencies and overly wordy sentences, and have made revisions throughout to ensure consistent past tense usage and improved readability.

(16) The sentences on lines 111-113 are quite wordy. The core conclusion, which is that the protein affects the insect's feeding probe, could be expressed more simply and directly to improve clarity and flow. I suggest rephrasing this section to be more concise and to highlight the primary finding without the added language.

We thank the reviewer for the helpful suggestion. We have revised the sentences to make them more concise and to emphasize the main finding that BtRDP influences the whitefly's feeding behavior as follow: "Compared with the dsGFP control, dsBtRDP-treated *B. tabaci* showed a marked reduction in phloem ingestion and a longer pathway duration, indicating that BtRDP is required for efficient feeding (Fig. 2c)."

(17) On line 118, the authors mention "subcellular location." It is not clear where the protein is localized. The authors should explicitly state the specific subcellular compartment of the protein, as this is crucial for understanding its function and interaction with other proteins.

We thank the reviewer for this valuable comment. To clarify the subcellular localization of BtRDP, we have revised the manuscript accordingly. The transgenic line overexpressing the full-length BtRDP including the signal peptide (oeBtRDP) is expected to localize in the apoplast (extracellular space), whereas the line expressing BtRDP without the signal peptide (oeBtRDP^{SP}) is likely retained in the cytoplasm.

(18) Lines 121-128, the description of the fecundity and choice assays in this section is overly wordy. The authors should present the main conclusion of these experiments more directly and concisely. The key finding is that the protein affects feeding behavior; this central point is somewhat lost in the detailed, and sometimes repetitive, phrasing.

We thank the reviewer for this suggestion. In the revised manuscript, we have simplified the description of the fecundity and two-choice assays to highlight the main conclusion as follow: "Fecundity and two-choice assays showed that BtRDP, whether localized in the apoplast (oeBtRDP) or cytoplasm (oeBtRDP^{SP}), enhanced whitefly settling and oviposition compared with EV controls (Fig. 2d-i; Fig. S10), indicating that BtRDP promotes whitefly feeding behavior regardless of its subcellular location."

(19) Line 148, the manuscript mentions experiments involving transformation, but the transformation efficiency is not provided. Please include the transformation efficiency for all transformation experiments, as this is crucial for the reproducibility of the results.

We thank the reviewer for raising this point. We would like to clarify that no transformation experiments were performed in this section. The experiments described involved Y2H screening using BtRDP^{SP} as a bait to identify interacting proteins from a *N. benthamiana* cDNA library. Therefore, there is no transformation efficiency to report.

(20) Line 159, the manuscript refers to a sequence similarity around line 159 but does not provide the specific data. It is important to show the actual sequence similarity, perhaps in a supplementary figure or table, to support the claims being made.

We thank the reviewer for this suggestion. To support our statement regarding sequence similarity, we have added the corresponding alignment figure in the Fig. S11.

(21) Line 159, the manuscript refers to "three randomly selected salivary proteins." It is unclear from where these proteins were selected. The authors should clarify the source of this selection (e.g., a specific database or a previous study) to ensure the methodology is transparent and the results are reproducible.

We thank the reviewer for raising this point. These proteins were selected based on previously reports (10.1093/molbev/msad221; 10.1111/1744-7917.12856). In the current version, we provide the accession of these proteins in the MS.

(22) Line 160, the description "NtcCf9 without signal peptide and transmembrane domains" is difficult to understand. It would be clearer and more consistent to use a term like "truncated NtcCf9" and then specify which domains were removed, as this is a standard practice in molecular biology for describing protein constructs.

We thank the reviewer for this suggestion. We have revised the manuscript to describe the construct as "truncated NtCf9" and specified that the signal peptide and transmembrane domains were removed

(23) The phrase "incubated with anti-flag beads" on line 172 is a detail of a routine method. Such details are more appropriate for the Methods section rather than the main text, which should focus on the results and their implications. Please remove such descriptions from the main text to improve readability and flow.

We thank the reviewer for this suggestion. We have removed the methodological detail from the main text to improve readability. We also check this throughout the MS.

I am excited about the potential of this work and look forward to seeing the current version.

We sincerely thank the reviewer for the positive feedback and encouragement. We appreciate your time and thoughtful comments.

Reviewer #2 (Public review):

Summary:

The authors tested an interesting hypothesis that white flies and planthoppers independently evolved salivary proteins to dampen plant immunity by targeting a receptor-like protein.

Strengths:

The authors used a wide range of methods to dissect the function of the white fly protein BtRDP and identify its host target NtRPL4.

Thank you very much for your comments. We have carefully revised the MS following your valuable suggestions and comments.

Weaknesses:

(1) Serious concerns about protein work.

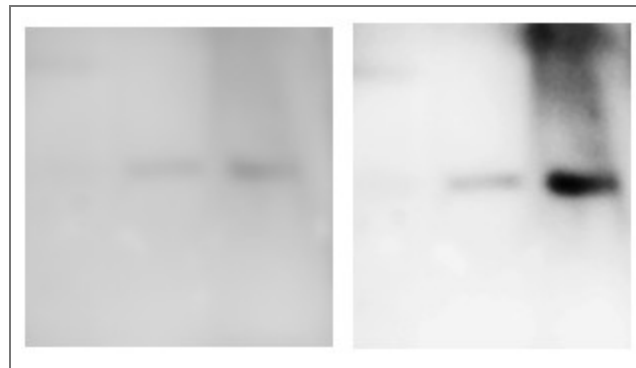
I did not find the indicated protein bands for anti-BtRDP in Figures 1a and 1b in the original blot pictures shown in Figure S30. In Figure 1a, I can't get the point of showing an unspecific protein band with a size of ~190 kD as a loading control for a protein of ~30 kD.

The data discrepancy led me to check other Western blot pictures. Similarly, Figures 2d, 3b, 3d, and S15b (anti-Myc) do not correspond to the original blots shown. In addition, the anti-Myc blot in Figure 4i, all blot pictures in Figures 5b, 5h, and S19a appeared to be compressed vertically. These data raised concerns about the quality of the manuscript.

Blots shown in Figure 3d, 4f, 4g, and 4h appeared to be done at a different exposure rate compared to the complete blot shown in Figure S30. The undesirable connection between Western blot pictures shown in the figures and the original data might be due to the reduced quality of compressed figures during submission. Nevertheless, clarification will be necessary to support the strength of the data provided.

We sincerely thank the reviewer for carefully examining our Western blot data and for pointing out these inconsistencies. The discrepancy between the figures in the main text and the original blots (Figure S30) resulted from an oversight during manuscript revision. This manuscript had undergone multiple rounds of revision after submission to another journal. During this process, the main figures and supplementary figures were updated separately, and we mistakenly failed to replace the original blot files with the corresponding current versions.

For the different exposure rate, the blots shown in the main text were adjusted for overall contrast and brightness to enhance band visibility and presentation clarity, whereas the original images in Figure S30 were raw, unprocessed scans directly from the imaging system. For example, in the Author response image 1 below, to visualize the loading of the input sample, the output figure was adjusted for overall contrast and brightness. This was acceptable for image processing (<https://www.nature.com/nature-portfolio/editorial-policies/image-integrity>)



Author response image 1. The same figure with brightness and contrast changes across the entire image.

For the vertical compression, in the previous version, some images were vertically compressed for layout purposes to make the composite figures appear more visually balanced. However, after consulting relevant publication guidelines, we realized that such one-dimensional compression is not encouraged by certain journals as it may alter the original aspect ratio of the image. Therefore, in the manuscript, we have avoided any non-proportional scaling and retained the original aspect ratio of all images.

We have now carefully rechecked all Western blot data, replaced the outdated raw blot images with the correct corresponding ones, avoid vertical compression, and ensured that the processed figures in the main text match their original data. The revised supplementary figures now accurately reflect the raw experimental results.

(2) *Misinterpretation of data.*

*I am afraid the authors misunderstood pattern-triggered immunity through receptor-like proteins. It is true that several LRR-type RLPs constitutively associate with SOBIR1, and further recruit BAK1 or other SERKs upon ligand binding. One should not take it for granted that every RLP works this way. To test the hypothesis that NtRLP4 confers resistance to *B. tabaci* infestation, the author compared transcriptional profiles between an EV plant line and an RLP4 overexpression line. If I understood the methods and figure legends correctly, this was done without *B. tabaci* treatment. This experimental design is seriously flawed. To provide convincing genetic evidence, independent mutant lines (optionally independent overexpression lines) in combination with different treatments will be necessary. Otherwise, one can only conclude that overexpressing the RLP4 protein generated a nervous plant. In addition, ROS burst, but not H₂O₂ accumulation, is a common immune response in pattern-triggered immunity.*

We agree with the reviewer that not every RLP functions through the same mechanism as the canonical SOBIR1–BAK1 pathway. In the current version, we further examined the interaction between the whitefly salivary protein and SOBIR1, and found that they do not interact. However, our interaction assays clearly demonstrated that NtRLP4 does interact with SOBIR1. Whether NtRLP4 functions through, or exclusively through, SOBIR1 remains uncertain, and we have emphasized this limitation in the Discussion section as follows: “Although NtRLP4 interacts with SOBIR1, this alone does not confirm that it operates strictly through this canonical module. Evidence from other RLPs shows that co-receptor usage can be flexible, and some RLPs function partly or conditionally independent of SOBIR1 [39]. Therefore, a more definitive assessment of NtRLP4 signaling will therefore require genetic dissection of its co-receptor dependencies, including but not limited to SOBIR1.”

Regarding the transcriptome analysis, our original aim was to explore why *B. tabaci* showed such a pronounced preference among tobacco plants. As this preference was assessed using uninfested plants, we also performed transcriptome sequencing using plants without *B. tabaci* treatment. The enrichment analysis demonstrated that the majority of up-regulated DEGs were associated with plant–pathogen interaction, environmental adaptation, MAPK signaling, and signal transduction pathways, while down-regulated DEGs were enriched in glutathione, carbohydrate, and amino acid metabolism. Notably, many DEGs were annotated as RLK/RLPs or WRKY transcription factors, most of which were upregulated, suggesting an enhanced defense state in the NtRLP4-overexpressing plants. The altered expression of JA- and SA-related genes (e.g., upregulation of FAD7 and downregulation of PAL and NPR1) further supported this enhanced defense and hormonal crosstalk. We agree that combining overexpression or knockout lines with insect infestation treatments would provide more direct genetic evidence for NtRLP4-mediated resistance, and we have acknowledged this as an important future direction. Nevertheless, our current data are consistent with the conclusion that NtRLP4 overexpression confers increased resistance to *B. tabaci* infestation.

Finally, DAB staining for H₂O₂ accumulation is also a well-established indicator of PTI responses, and many studies have shown that overexpression of salivary elicitors can trigger such accumulation.

(3) Lack of logic coherence.

The written language needs substantial improvement. This impeded the readability of the work. More importantly, the logic throughout the manuscript appeared scattered. The choice of testing protein domains for protein-protein interactions, using plants overexpressing an insect protein to study its subcellular localization, switching back and forth between using proteins with signal peptides and without signal peptides, among others, lacks a clear explanation.

We appreciate the reviewer's careful reading and valuable comments regarding the logical coherence of our manuscript.

(1) To improve the English quality, the entire manuscript has been professionally edited by a certified language-editing service.

(2) Regarding the rationale for testing protein domains in the protein–protein interaction assays: NtRLP4 is a membrane-anchored receptor-like protein composed of extracellular, transmembrane, and short intracellular domains. We aimed to determine which region of NtRLP4 is responsible for interacting with the salivary protein, as this would help infer the likely site of interaction in planta. In addition, not all RLPs contain a malectin-like domain, and we sought to verify whether the BtRDP–NtRLP4 interaction depends on this domain. To enhance the logical flow, we introduced a brief statement explaining the experimental purpose before presenting the interaction assays in the current version as follow: “These findings raised the question of which domain of NtRLP4 is responsible for binding BtRDP, as identifying the interacting domain could help infer where the salivary protein contacts the receptor in planta. We therefore dissected the NtRLP4 domains accordingly.”

(3) With respect to using plants overexpressing an insect protein to examine subcellular localization: since both the brown planthopper and the whitefly are non-model species for which stable genetic transformation is technically unfeasible, many previous studies have used *Agrobacterium*-mediated transient expression or transgenic plant systems to investigate the subcellular localization of insect salivary proteins within host cells. Following these precedents, our study also employed plant systems to determine the localization of the insect protein and to assess how different localizations affect plant defense responses.

(4) As for switching between constructs with or without signal peptides: the subcellular localization of effectors can influence their biological activity and interactions. Previous studies have used the presence or absence of signal peptides, or replacement with a PR1 signal peptide, to direct protein targeting (for example, *Frontiers in Plant Science*, 2022, 13:813181). Because salivary sheaths are generally considered to localize in the apoplastic space, we generated two transgenic *N. tabacum* lines overexpressing BtRDP: one carrying the full-length coding sequence including the signal peptide (oeBtRDP), expected to be secreted into the apoplast, and another lacking the signal peptide (oeBtRDP-sp), likely retained in the cytoplasm. In the current version, we clarified this rationale and added references to similar studies to improve the manuscript's logic and readability. Details are as follow: “To investigate the role of BtRDP in different subcellular location of host plants, we constructed two transgenic *N. tabacum* lines overexpressing BtRDP: one carrying the full-length coding sequence including the signal peptide (oeBtRDP), which is expected to be secreted into the apoplast (extracellular space), and the other lacking the signal peptide (oeBtRDP^{sp}), which is likely retained in the cytoplasm.”

Reviewer #3 (Public review):

Summary:

In this study, Wang et al. investigate how herbivorous insects overcome plant receptor-mediated immunity by targeting plant receptor-like proteins. The authors identify two independently evolved salivary effectors, BtRDP in whiteflies and NISP694 in brown planthoppers, that promote the degradation of plant RLP4 through the ubiquitin-dependent proteasome pathway. NtRLP4 from tobacco and OsRLP4 from rice are shown to confer resistance against herbivores by activating defense signaling, while BtRDP and NISP694 suppress these defenses by destabilizing RLP4 proteins.

Strengths:

This work highlights a convergent evolutionary strategy in distinct insect lineages and advances our understanding of insect-plant coevolution at the molecular level.

Thank you very much for your comments. We have carefully revised the MS following your valuable suggestions and comments.

Weaknesses:

(1) I found the naming of BtRDP and NISP694 somewhat confusing. The authors defined BtRDP as "B. tabaci RLP-degrading protein," whereas NISP694 appears to have been named after the last three digits of its GenBank accession number (MF278694, presumably). Is there a standard convention for naming newly identified proteins, for example, based on functional motifs or sequence characteristics? As it stands, the inconsistency makes it difficult for readers to clearly distinguish these proteins from those reported in other studies.

Thank you for your comment. These are species-specific salivary proteins that have not been reported or annotated in previous studies. Because no homologous genes could be identified in other species, there are no existing names or annotations for these proteins. For such lineage-specific salivary proteins, it is common in recent studies to name them according to their experimentally identified functions. For example, a recently reported salivary protein was named SR45-interacting salivary protein (SISP) based on its function (10.1111/nph.70668). Following this convention, we adopted a similar functional naming strategy in this study. We acknowledge that there may not yet be a standardized rule for naming such proteins, and we would be glad to follow a more authoritative naming guideline if possible.

(2) Figure 2 and other figures. Transgenic experiments require at least two independent lines, because results from a single line may be confounded by position effects or unintended genomic alterations, and multiple lines provide stronger evidence for reproducibility and reliability.

We appreciate the reviewer's suggestion. In our study, two independent transgenic lines were used to ensure the reproducibility and reliability of the results. One representative line was presented in the main figures, while data from the second independent line were included in the supplementary figures. To make this clearer, we have emphasized in the manuscript that bioassays were conducted using two independent transgenic lines.

(3) Figure 3e. Quantitative analysis of NtRLP4 was required. Additionally, since only one band was observed in oeRLP, were any tags included in the construct?

Thank you for your comment. In the current version, quantitative analysis of NtRLP4 expression has been performed and is now presented in Figure 3. For the oeRLP plants, no tag was fused to NtRLP4; thus, anti-RLP serum was used to detect the target bands. In contrast, oeBtRDP and oeBtRDP-sp were fused with C-terminal FLAG tags, and their detection was carried out using anti-FLAG serum. This information has been clarified in the revised Methods section as follows: "The oeBtRDP and oeBtRDP^{sp} were fused with C-terminal FLAG tags, while no tag was fused to oeNtRLP4."

(4) Figure 4a. The RNAi effect appears to be well rescued in Line 1 but poorly in Line 2. Could the authors clarify the reason for this difference?

Thank you for pointing this out. We also noticed that the RNAi effect appeared to be better rescued in Line 2 than in Line 1. Based on our measurements, the silencing efficiency of NtRLP4 in RNAi-RLP4 Line 1 was markedly weaker than in Line 2, which likely explains the difference in rescue efficiency. In the current version, we have clarified this point as follows: "Both RNAi-RLP lines showed reduced NtRLP4 levels compared with EV plants, with RNAi-

RLP#2 exhibiting a stronger silencing effect (Fig. S19a).” “The differential rescue effect between the two RNAi lines likely resulted from their different NtRLP4 silencing efficiencies, with the lower NtRLP4 level in RNAi-RLP#2 leading to a more complete rescue phenotype.”

(5) ROS accumulation is shown for only a single leaf. A quantitative analysis of ROS accumulation across multiple samples would be necessary to support the conclusion. The same applies to Figure 16f.

Thank you for pointing this out. The H₂O₂ accumulation experiments have been repeated for 5 times in Figure 4 and Figure S16f. In the current version, we addressed that “the experiment is repeated five times with similar results” in the figure legends.

(6) Figure 4f: NtRLP4 abundance was significantly reduced in oeBtRDP plants but not in oeBtRDP-SP. Although coexpression analysis suggests that BtRDP promotes NtRLP4 degradation in an ubiquitin-dependent manner, the reduced NtRLP4 levels may not result from a direct interaction between BtRDP and NtRLP4. It is possible that BtRDP influences other factors that indirectly affect NtRLP4 abundance. The authors should discuss this possibility.

Thank you for your valuable suggestion. We agree that the reduced NtRLP4 abundance may not necessarily result from a direct interaction between BtRDP and NtRLP4. In the manuscript, we have further discussed this possibility as follows: “Notably, BtRDP and NISP104 shared no sequence or structural similarity and lack resemblance to known eukaryotic ubiquitin-ligase domains. Their interaction with RLP4s occurs in the extracellular space (Fig. 3d; Fig. 5c), whereas the ubiquitin-proteasome system primarily functions in the cytosol and nucleus [46]. Furthermore, NtRLP4 reduction is observed only in oeBtRDP transgenic plants, not in oeBtRDP-sp plants (Fig. 4f), suggesting that BtRDP exerts its influence on NtRLP4 in the extracellular space. These observations collectively argue against the possibility that BtRDP or NISP694 possesses intrinsic E3 ligase activity capable of directly ubiquitinating RLP4s within plant cells. Importantly, the reduced NtRLP4 levels may not result from a direct physical interaction between BtRDP and NtRLP4. Instead, BtRDP may indirectly affect RLP4 post-translational modification, thereby accelerating its degradation, which warrants further investigation”

(7) The statement in lines 335-336 that 'Overexpression of NtRLP4 or NtSOBIR1 enhances insect feeding, while silencing of either gene exerts the opposite effect' is not supported by the results shown in Figures S16-S19. The authors should revise this description to accurately reflect the data.

Thank you for pointing this out. We agree that our original statement was not precise, as we measured the insect settling preference and oviposition on transgenic plants, but did not directly assess the feeding behavior of *B. tabaci*. Therefore, we have revised the description in the manuscript to more accurately reflect our data as follows: “Overexpression of NtRLP4 or NtSOBIR1 in *N. tabacum* is attractive to *B. tabaci* and promotes insect reproduction, whereas silencing of either gene exerts the opposite effect.”

(8) BtRDP is reported to attach to the salivary sheath. Does the planthopper NISP694 exhibit a similar secretion localization (e.g., attachment to the salivary sheath)? The authors should supplement this information or discuss the potential implications of any differences in secretion localization between BtRDP and NISP694 for their respective modes of action.

Thank you for your insightful suggestion. We agree that determining the secretion localization of NISP694 would provide valuable information for understanding its potential mode of action. Immunohistochemical (IHC) staining is indeed a critical approach for such analysis. However, in this study, we were unable to express NISP694 in *Escherichia coli*, and the antibody generated using a synthesized peptide did not show sufficient specificity or

sensitivity for IHC detection. Consequently, we were unable to determine whether NISP694 is attached to the salivary sheath. Therefore, whether BtRDP and NISP694 acted in different mode require further investigation.

Recommendations for the authors:

Reviewer #3 (Recommendations for the authors):

(1) Figure 1e. The BtRDP-labeled fluorescent signal is difficult to discern. An enlarged view of the target region would be helpful for clarity.

Thank you for your suggestion. In the current version, an enlarged view of the target region was provided below the figure.

(2) The finding that BtRDP accumulates in the salivary sheath secreted by Bemisia tabaci is important for understanding the subcellular localization of this protein during actual insect feeding. I suggest moving Figure S5 to the main text.

Thank you for your suggestion. Figure S5 has been moved to Fig. 1f in the current version.

(3) Please carefully cross-check the figure numbering to ensure that all in-text citations correspond to the correct figures and panels. i.e., lines 136,188,192, and 194.

Thank you for pointing this out. We corrected them in the current version.

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